
DO AI WEATHER MODELS MISS EXTREMES? EVIDENCE FROM TEN MONTHS OF VERIFICATION AGAINST EUROPEAN WEATHER STATIONS

Anonymous authors

Paper under double-blind review

ABSTRACT

AI weather models are often reported to underestimate extremes, but most evidence concerns deterministic regression models verified against reanalysis. We evaluate twelve physical and AI forecast models against ECMWF IFS using ten months of European station observations. The evaluation covers 10 m wind, 2 m temperature, solar radiation, and precipitation within regimes defined from a fixed ERA5 1991–2020 climatology. We find no uniform AI-specific deficit in the tails. Several AI models remain more accurate than IFS under extreme conditions, while others deteriorate markedly; comparable variation occurs among physical models. Every model nevertheless exhibits a common conditional-error pattern, overpredicting low observations and underpredicting high observations. Attenuation of extreme values therefore does not imply a uniform loss of relative skill: tail performance depends on the model, variable, and evaluation setting rather than on whether the forecast is produced by AI or physical numerical modelling.

1 INTRODUCTION

Data-driven weather models now outperform operational physical models on many standard forecast metrics (Bi et al., 2023; Lam et al., 2023; Ben Bouallègue et al., 2024). Their performance at extremes remains less certain. Models trained with mean-squared error tend towards the conditional mean, potentially smoothing sharp gradients and attenuating rare events (Olivetti & Messori, 2024a; Bonavita, 2024). Evaluations of early AI weather models accordingly report underestimated peak winds, hot and windy tails, and record-breaking events (Charlton-Perez et al., 2024; Olivetti & Messori, 2024b; Zhang et al., 2026).

Most of this evidence concerns deterministic regression models evaluated against reanalysis. Generative ensembles, probabilistically trained models, and high-resolution regional AI models have received less attention, particularly against direct observations. This distinction matters because station-based errors can be substantially larger than reanalysis-based estimates (Jin et al., 2024; Gupta et al., 2025; Schlueter & Wagner, 2025).

Recent work also suggests that attenuation is not intrinsic to data-driven forecasting. Grid-point regression objectives can reduce effective resolution through the double penalty, whereas amplitude-preserving losses improve surface-wind extremes (Subich et al., 2025). Diffusion models preserve flow statistics that regression ensembles tend to suppress (Molinaro et al., 2024), and generative weather models have demonstrated skill for reanalysis-defined extremes and convection-allowing regional forecasting (Price et al., 2025; Pathak et al., 2026). Direct station-based comparisons across these model classes nevertheless remain limited.

Two questions are often conflated: whether a forecast attenuates an extreme observation and whether it is less accurate than a competing model under the same conditions. The first concerns absolute error structure; the second concerns relative skill. A model may underestimate an extreme while remaining closer to the observation than its physical counterpart.

We examine this distinction using thirteen forecast models, including ECMWF IFS as the reference. Forecasts are verified over ten months against European synoptic, solar-radiation, and rain-gauge

observations. The evaluation covers 10 m wind, 2 m temperature, hourly solar radiation, and hourly precipitation at lead times up to 48 h.

Throughout the paper, “extreme” denotes the tails of a local ERA5 1991–2020 climatology conditioned on station, calendar day, and UTC hour. These regimes identify locally unusual conditions rather than necessarily record-breaking or impact-defined events. We compare relative MAE within these regimes with the corresponding conditional forecast errors. This separates model-specific skill from the common tendency of forecasts to overpredict low observations and underpredict high observations. The study makes three contributions:

- We extend station-based evaluation of weather extremes to a diverse set of forecasting approaches, including deterministic regression models, a CRPS-trained global ensemble, and high-resolution regional generative models.
- We disentangle model performance across climatological regimes from the conditional forecast errors within those regimes, separating differences in regime frequency from differences in forecast skill.
- We assess the robustness of these patterns across four surface variables, multiple forecast leads, countries, and model classes, and further evaluate heavy precipitation as a categorical extreme.

2 METHODS

2.1 MODELS

We evaluate twelve forecast models against the ECMWF IFS high-resolution deterministic forecast, which serves as the reference (Table S1). The comparison spans physical and AI models, global and regional domains, and deterministic and ensemble forecasts.

The physical models are ECMWF ENS, NOAA GFS, DWD ICON Global, and DWD ICON-EU. The AI models comprise the global regression models ECMWF AIFS, Microsoft Aurora, and Jua EPT-2 Reasoning; the global ensembles ECMWF AIFS ENS and Jua EPT-2e; the regional generative ensembles Jua EPT-2 HRRR and EPT-2.1 Europa; and the solar-specialised regional model EPT-2.1 Helios (Lang et al., 2024a; Bodnar et al., 2025; Molinaro et al., 2025; Lang et al., 2024b; Siegenheim, 2025; Jua AI, 2026; Molinaro, 2026). AIFS ENS is trained with a continuous ranked probability score objective and learned noise injection and is evaluated as a 51-member ensemble. Related generative methodology is described by Molinaro et al. (2026; 2024).

Models are evaluated only for variables with compatible outputs. Precipitation is available from the physical models and from EPT-2 Reasoning, EPT-2e, EPT-2 HRRR, and EPT-2.1 Europa. AIFS ENS is restricted to wind and temperature because its radiation and precipitation outputs cannot be matched to the hourly verification data.

2.2 STATION OBSERVATIONS AND EVALUATION

Forecasts are verified against three complementary in situ networks: a synoptic network for 10 m wind and 2 m temperature, a solar network for hourly global horizontal irradiance, and a panel of 4,005 rain gauges for hourly precipitation. The separate gauge network is required because the synoptic feed does not provide usable hourly precipitation accumulations. Network providers are listed in Appendix A.

The evaluation covers thirteen countries: DE, FR, GB, ES, IT, PL, NL, BE, CH, NO, AT, CZ, and DK. The precipitation network excludes Italy and Poland. Forecasts are bilinearly interpolated to observation locations (Molinaro et al., 2026). Precipitation is scored only at gauges within each model’s domain, and regional comparisons use a common gauge sample. None of the Jua models was trained or fine-tuned on these observations; historical station errors enter only through the causal bias correction described below.

Two evaluation tracks account for differences in availability and output cadence. Track A covers 1 September 2025 to 30 June 2026. Wind and temperature use 6-hourly leads from 6 to 48 h, matching the cadence of AIFS, AIFS ENS, and Aurora. Solar radiation and hourly precipitation use leads

from 1 to 48 h. Models without compatible outputs are omitted for the corresponding variable, and ECMWF ENS contributes to precipitation only at its native 6-hourly leads.

Track B covers 1 March to 30 June 2026 and compares EPT-2.1 Europa, EPT-2 HRRR, and ICON-EU over a common hourly grid from 1 to 48 h. EPT-2.1 Helios is additionally included for solar radiation. Detailed availability by variable, country, and period is reported in the supplement.

2.3 OBSERVED-VALUE REGIMES

Regimes are defined independently of the evaluation period using the fixed ERA5 1991–2020 climatology (Hersbach et al., 2020). ERA5 fields are bilinearly interpolated from the 0.25° grid to each station. For every station, calendar day, and UTC hour, the reference distribution contains values from a cyclic ± 15 -day window over the 30-year period, yielding 930 values before variable-specific conditioning.

We denote percentiles of this distribution by P5, P25, P50, P75, and P95. Wind and temperature use all climatological values, while solar percentiles are estimated from positive daytime irradiance. Precipitation below 0.1 mm forms a separate dry regime; for precipitation, P50, P75, and P95 always denote percentiles of wet hours. Each forecast–observation pair is assigned to a regime according to its observed value y , so regime membership is identical across models and leads. The complete definitions and meteorological labels are given in Tables S2 and S3.

This local-climatology construction follows established practice for climate extremes (Zhang et al., 2011; Klein Tank et al., 2009; Perkins & Alexander, 2013). Because ERA5 represents grid-cell averages, its variance is lower than that of point observations; observed wind and solar tails are consequently broader than their nominal climatological probabilities.

2.4 BIAS CORRECTION

We apply a causal rolling bias correction before computing skill. For wind, temperature, and daytime solar radiation, the correction is additive. Let $\mathcal{D}_{w,h}$ contain the matched errors at valid hour h during the four weeks preceding week w . For model m ,

$$\bar{b}_{m,w,h} = \frac{1}{|\mathcal{D}_{w,h}|} \sum_{(f_m,y) \in \mathcal{D}_{w,h}} (f_m - y), \quad \tilde{f}_m(t) = f_m(t) - \bar{b}_{m,w,h(t)}. \quad (1)$$

The estimate is pooled across European stations and does not depend on initialisation or lead. For ensemble models, the same shift is applied to every member.

An additive correction is unsuitable for zero-bounded, right-skewed precipitation. Let $\mathcal{P}_w$ contain forecasts and observations from the preceding four weeks, pooled across initialisations and leads. We apply

$$a_{m,w} = \text{clip}_{[0.33, 3]} \left(\frac{\sum_{\mathcal{P}_w} y}{\sum_{\mathcal{P}_w} f_m} \right), \quad \tilde{f}_m(t) = a_{m,w} f_m(t). \quad (2)$$

This multiplicative correction adjusts systematic amount biases while preserving zero forecasts. Both procedures use only prior information and do not alter observation-defined regime membership or force the conditional error within each regime to zero.

2.5 SKILL AND AGGREGATION

For model m and evaluation cell c , MAE skill relative to ECMWF IFS is

$$S_{m,c} = 1 - \frac{\text{MAE}_{m,c}}{\text{MAE}_{\text{IFS},c}}. \quad (3)$$

Positive values indicate lower MAE than IFS. Scores are computed from matched forecast–observation pairs using the common intersection of available initialisations and the four daily cycles at 00, 06, 12, and 18 UTC.

Ensemble models are evaluated through their ensemble means; the analysis therefore assesses point-forecast accuracy rather than calibration or spread. Results are reported at selected leads and over

the complete lead grid of each comparison. Lower-cadence models contribute only at their native leads, without temporal interpolation.

Uncertainty is estimated using a leave-one-month-out jackknife and reported as one standard error. For exceedance of P95, we additionally report probability of detection, false-alarm ratio, critical success index, and frequency bias, pooled across gauges and leads up to 48 h.

3 RESULTS

3.1 SKILL ACROSS THE OBSERVED DISTRIBUTION

Figure 1 shows that tail skill does not follow a uniform AI–physical ordering. EPT-2.1 Europa leads the wind comparison overall and remains competitive with ICON Global at gale force. EPT-2 HRRR and EPT-2.1 Europa perform most strongly for temperature, including in the heat regime. By contrast, AIFS and AIFS ENS remain below IFS in the wind and temperature tails despite the ensemble’s improved performance near the centre of the distribution. The largest heat deficit belongs to the physical GFS model.

Solar radiation follows a different pattern. EPT-2.1 Helios leads overall and under overcast and near-clear-sky conditions but falls below IFS under typical irradiance, where EPT-2 Reasoning performs most strongly. Its advantage is therefore concentrated in the tails rather than distributed uniformly across conditions.

For precipitation, the clearest gains occur in the middle of the wet-hour distribution (Figure 2). EPT-2 HRRR, EPT-2e, and EPT-2 Reasoning improve on IFS from moderate through high rain, whereas the physical global models remain below the reference. These gains narrow at the heaviest intensities. Europa follows a distinct pattern, deriving much of its aggregate skill from dry hours while losing skill once precipitation occurs.

3.2 SHARED CONDITIONAL-ERROR PATTERN

Despite their differences in relative skill, all models exhibit the same conditional-error structure (Figure 3): low observations are overpredicted and high observations underpredicted. The variation across regimes is larger than the spread between models and is present for all four variables. A common tendency towards the centre of the observed distribution therefore coexists with model-specific differences in relative accuracy.

Because regimes are conditioned on the verifying observation, this pattern is not by itself evidence of smoothing. Limited predictability, unresolved local variability, and observational or representativeness errors can produce the same structure.

3.3 REGIONAL COMPARISON

Track B compares EPT-2.1 Europa and EPT-2 HRRR directly with ICON-EU over March–June 2026 (Figure 4). Europa retains the strongest overall wind estimate, while ICON-EU is competitive with the leading AI models in the wind and temperature tails. EPT-2.1 Helios has the clearest solar advantage.

Supplementary analyses show that these conclusions persist across lead times, short-range forecasts, and individual countries (Appendix A), which also reports complete precipitation, uncertainty, and categorical-verification results.

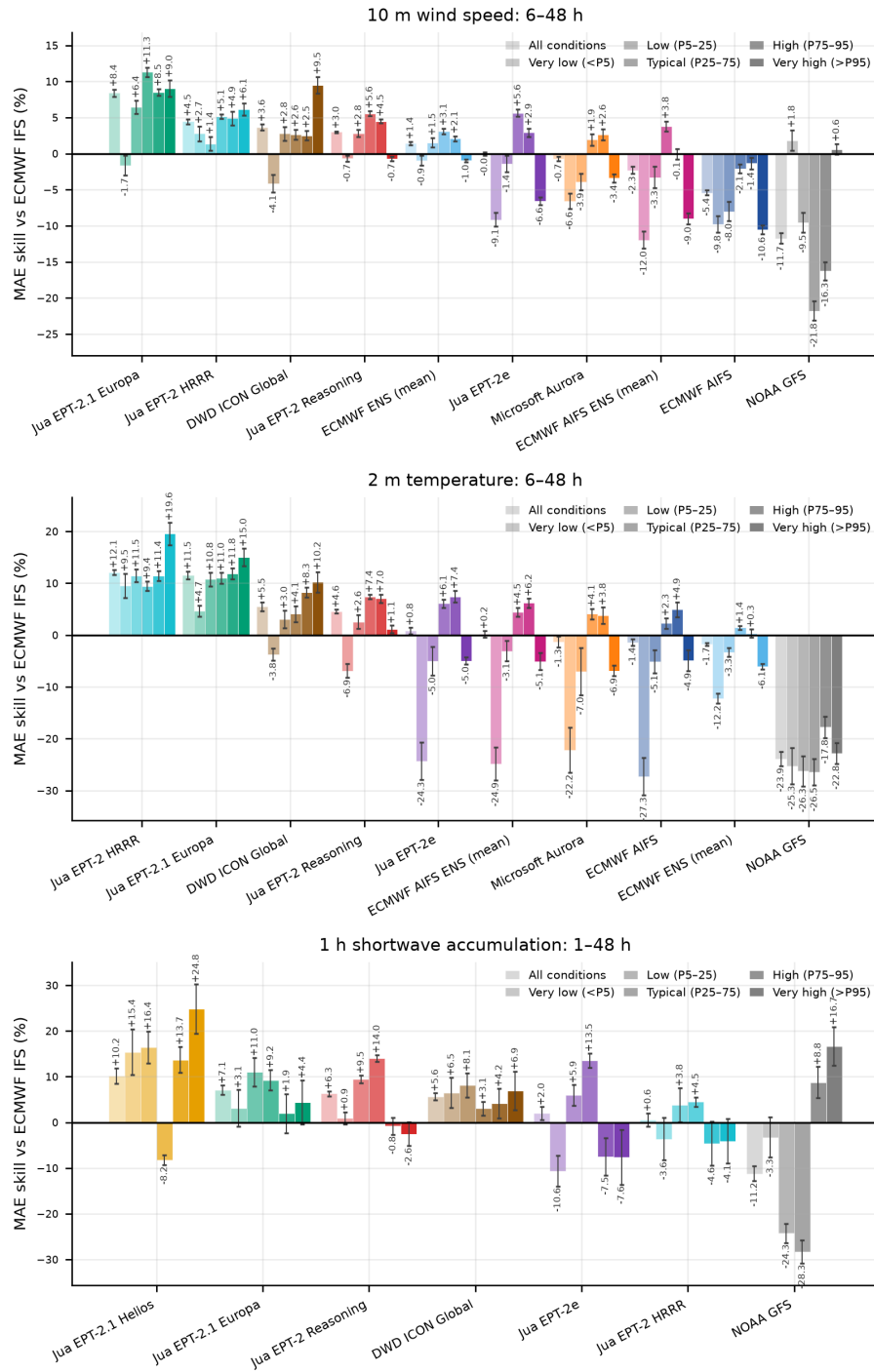


Figure 1: Track A MAE skill relative to ECMWF IFS by observed-value regime, with ± 1 jackknife SE. Wind and temperature use 6–48 h leads; solar radiation uses 1–48 h.

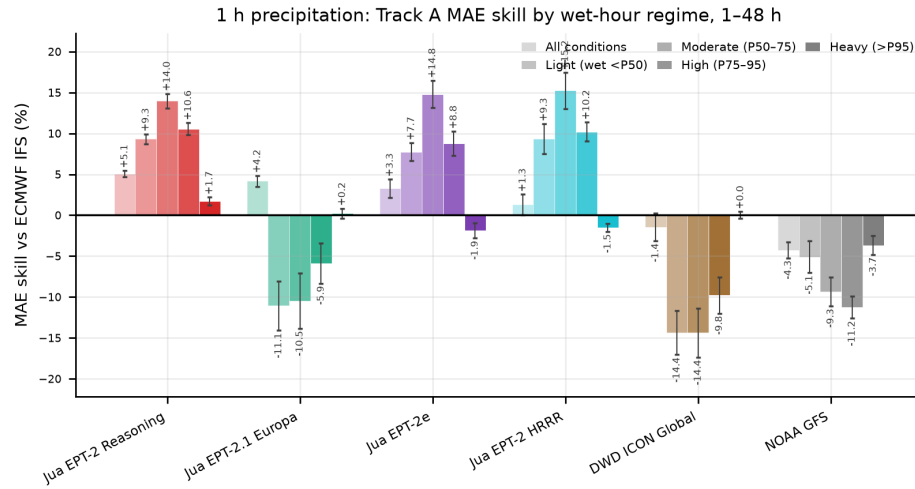


Figure 2: Track A MAE skill relative to ECMWF IFS for hourly precipitation by observed-value regime over 1–48 h, with ± 1 jackknife SE.

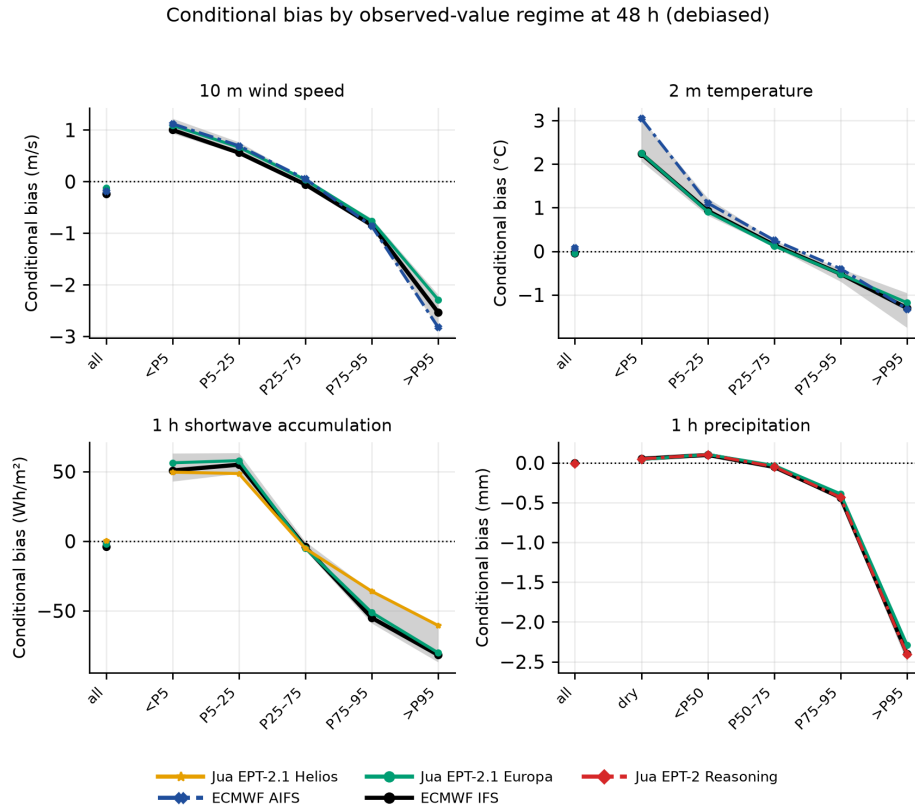


Figure 3: Mean forecast-minus-observation error at 48 h, conditioned on the observed-value regime after bias correction. The grey envelope spans all models; selected models are highlighted.

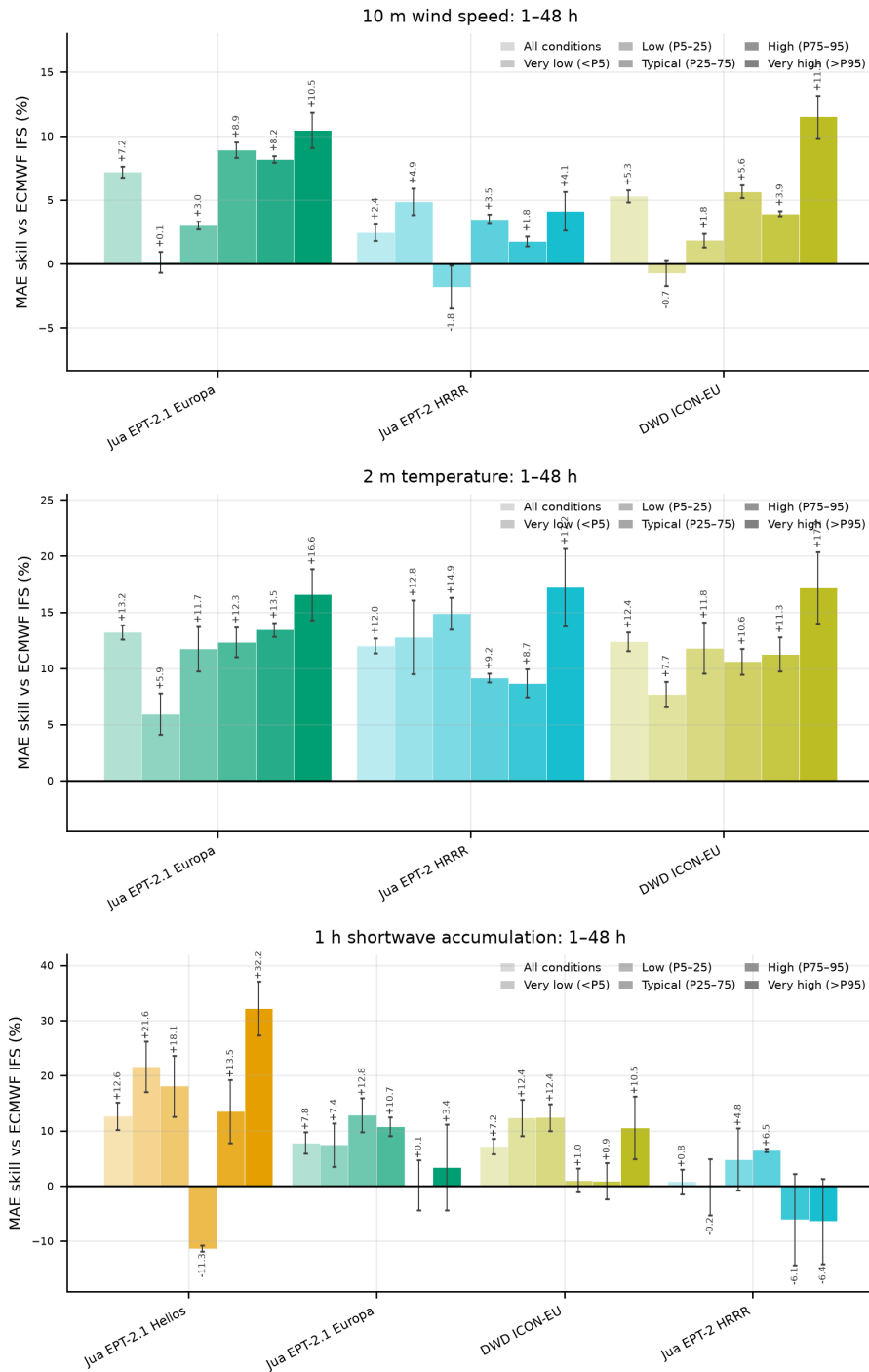


Figure 4: Track B MAE skill relative to ECMWF IFS by observed-value regime, aggregated over 1–48 h from March to June 2026, with ± 1 jackknife SE. Wind and temperature compare EPT-2.1 Europa, EPT-2 HRRR, and ICON-EU; the solar panel additionally includes EPT-2.1 Helios. Models are ordered by all-conditions skill within each panel.

4 DISCUSSION

The results distinguish two meanings of “missing extremes.” A forecast may underestimate an extreme observation, or it may be less accurate than a competing model under the same conditions. The first behaviour is common: all evaluated models tend to overpredict low observations and underpredict high observations. The second is not. Several AI models achieve lower MAE than IFS in the heat, gale, solar, and precipitation regimes. Attenuation of extreme values therefore does not imply a general AI-specific deficit in relative skill.

This distinction qualifies earlier findings based mainly on deterministic regression models evaluated against reanalysis (Olivetti & Messori, 2024b). The broader station-based comparison reveals substantial variation across models and variables. EPT-2.1 Helios performs strongly in both solar tails but loses skill under typical irradiance, while the precipitation gains of the Jua models are concentrated in the middle of the wet distribution. Tail performance is therefore not determined by aggregate skill alone.

The results are also inconsistent with a simple AI–physical divide. The wind- and temperature-tail deficits of AIFS persist for AIFS ENS despite its improved central-regime performance. Conversely, EPT-2 HRRR and EPT-2.1 Europa remain competitive at gale force and in heat. Physical models show comparable variation: GFS has the largest heat deficit, while ICON-EU is competitive with the leading regional AI models.

These comparisons do not establish the causes of the differences. The models vary in architecture, objective, resolution, training data, initialisation, cadence, and domain (Bonavita, 2024; Subich et al., 2025). Their performance cannot therefore be attributed independently to AI versus physics, regression versus generative training, or global versus regional resolution. Evaluating ensembles through their means also provides no evidence about calibration, spread, or extreme-event probabilities.

Absolute attenuation and relative skill are therefore complementary. AI models may underestimate extreme observations without being uniformly less accurate than physical models, while strong aggregate performance does not guarantee an advantage in every tail. Within this evaluation, performance at extremes is model- and variable-specific rather than a general weakness of AI weather forecasting.

ETHICS STATEMENT

Several of the forecast systems evaluated in this study are developed by the authors’ organisation. The verification protocol is applied identically across AI and physical systems, every comparison is reported against the independent ECMWF IFS reference, and the archived requests, aggregates and reproduction code described below are provided to support independent scrutiny of the comparisons. The study uses forecast output and meteorological station observations only; it involves no human subjects or personal data, and no station identities or per-station series are released.

REPRODUCIBILITY STATEMENT

Metrics were computed through a content-addressed station-verification API; requests and responses are archived. The reproduction package includes the aggregates used for the reported numbers, headline skill tables, and the extraction, aggregation and figure code. All score-based figures and tables regenerate offline from those aggregates. The package is provided as anonymised supplementary material.

AI USE STATEMENT

Generative AI tools were used to assist with software changes in the extraction, aggregation and figure code, to inspect generated results, and to draft and edit manuscript text. They were not used to generate or modify observations or forecast data. The authors checked AI-assisted code through the pipeline’s executable self-tests and regenerated artifacts, verified the quoted numbers against the aggregate data, reviewed all manuscript text, and take full responsibility for the content.

REFERENCES

- Zied Ben Bouallègue, Mariana C. A. Clare, Linus Magnusson, Estibaliz Gascón, Michael Maier-Gerber, Martin Janoušek, Mark Rodwell, Florian Pinault, Jesper S. Dramsch, Simon T. K. Lang, Baudouin Raoult, Florence Rabier, Matthieu Chevallier, Irina Sandu, Peter Dueben, Matthew Chantry, and Florian Pappenberger. The rise of data-driven weather forecasting: A first statistical assessment of machine learning-based weather forecasts in an operational-like context. *Bulletin of the American Meteorological Society*, 105(6):E864–E883, 2024. doi: 10.1175/BAMS-D-23-0162.1.
- Kaifeng Bi, Lingxi Xie, Hengheng Zhang, Xin Chen, Xiaotao Gu, and Qi Tian. Accurate medium-range global weather forecasting with 3D neural networks. *Nature*, 619(7970):533–538, 2023. doi: 10.1038/s41586-023-06185-3.
- Cristian Bodnar, Wessel P. Bruinsma, Ana Lucic, Megan Stanley, Anna Allen, Johannes Brandstetter, Patrick Garvan, Maik Riechert, Jonathan A. Weyn, Haiyu Dong, Jayesh K. Gupta, Kit Tham-biratnam, Alexander T. Archibald, Chun-Chieh Wu, Elizabeth Heider, Max Welling, Richard E. Turner, and Paris Perdikaris. A foundation model for the Earth system. *Nature*, 641(8065):1180–1187, 2025. doi: 10.1038/s41586-025-09005-y.
- Massimo Bonavita. On some limitations of current machine learning weather prediction models. *Geophysical Research Letters*, 51(12):e2023GL107377, 2024. doi: 10.1029/2023GL107377.
- Andrew J. Charlton-Perez, Helen F. Dacre, Simon Driscoll, Suzanne L. Gray, Ben Harvey, Natalie J. Harvey, Kieran M. R. Hunt, Robert W. Lee, Ranjini Swaminathan, Remy Vandaele, and Ambrogio Volonté. Do AI models produce better weather forecasts than physics-based models? A quantitative evaluation case study of Storm Ciarán. *npj Climate and Atmospheric Science*, 7: 93, 2024. doi: 10.1038/s41612-024-00638-w.
- Aman Gupta, Aditi Sheshadri, and Dhruv Suri. MAUSAM: An observations-focused assessment of global AI weather prediction models during the South Asian monsoon. arXiv preprint arXiv:2509.01879, 2025.
- Hans Hersbach, Bill Bell, Paul Berrisford, Shoji Hirahara, András Horányi, Joaquín Muñoz-Sabater, Julien Nicolas, Carole Peubey, Raluca Radu, Dinand Schepers, Adrian Simmons, Cornel Soci, Saleh Abdalla, Xavier Abellan, Gianpaolo Balsamo, Peter Bechtold, Gionata Biavati, Jean Bidlot, Massimo Bonavita, Giovanna De Chiara, Per Dahlgren, Dick Dee, Michail Diamantakis, Rossana Dragani, Johannes Flemming, Richard Forbes, Manuel Fuentes, Alan Geer, Leopold Haimberger, Sean Healy, Robin J. Hogan, Elías Hólm, Marta Janisková, Sarah Keeley, Patrick Laloyaux, Philippe Lopez, Cristina Lupu, Gabor Radnoti, Patricia de Rosnay, Iryna Rozum, Freja Vamborg, Sebastien Villaume, and Jean-Noël Thépaut. The ERA5 global reanalysis. *Quarterly Journal of the Royal Meteorological Society*, 146(730):1999–2049, 2020. doi: 10.1002/qj.3803.
- Weixin Jin, Jonathan Weyn, Pengcheng Zhao, Siqi Xiang, Jiang Bian, Zuliang Fang, Haiyu Dong, Hongyu Sun, Kit Thamiratnam, and Qi Zhang. WeatherReal: A benchmark based on in-situ observations for evaluating weather models. arXiv preprint arXiv:2409.09371, 2024.
- Jua AI. EPT-2 family: Models and products documentation. Jua documentation, <https://docs.jua.ai/models-and-products/jua-models/ept-2>, 2026. Accessed 18 July 2026; includes the EPT-2.1 Europa and EPT-2.1 Helios model specifications.
- Albert M. G. Klein Tank, Francis W. Zwiers, and Xuebin Zhang. Guidelines on analysis of extremes in a changing climate in support of informed decisions for adaptation. Technical Report WCDMP-No. 72, WMO-TD/No. 1500, World Meteorological Organization, Geneva, 2009.
- Remi Lam, Alvaro Sanchez-Gonzalez, Matthew Willson, Peter Wirnsberger, Meire Fortunato, Feran Alet, Suman Ravuri, Timo Ewalds, Zach Eaton-Rosen, Weihua Hu, Alexander Merose, Stephan Hoyer, George Holland, Oriol Vinyals, Jacklynn Stott, Alexander Pritzel, Shakir Mohamed, and Peter Battaglia. Learning skillful medium-range global weather forecasting. *Science*, 382(6677):1416–1421, 2023. doi: 10.1126/science.adi2336.

- Simon Lang, Mihai Alexe, Matthew Chantry, Jesper Dramsch, Florian Pinault, Baudouin Raoult, Mariana C. A. Clare, Christian Lessig, Michael Maier-Gerber, Linus Magnusson, Zied Ben Bouallègue, Ana Prieto Nemesio, Peter D. Dueben, Andrew Brown, Florian Pappenberger, and Florence Rabier. AIFS – ECMWF’s data-driven forecasting system. *arXiv preprint arXiv:2406.01465*, 2024a.
- Simon Lang, Mihai Alexe, Mariana C. A. Clare, Christopher Roberts, Rilwan Adewoyin, Zied Ben Bouallègue, Matthew Chantry, Jesper Dramsch, Peter D. Dueben, Sara Hahner, Pedro Maciel, Ana Prieto-Nemesio, Cathal O’Brien, Florian Pinault, Jan Polster, Baudouin Raoult, Steffen Tietsche, and Martin Leutbecher. AIFS-CRPS: Ensemble forecasting using a model trained with a loss function based on the Continuous Ranked Probability Score. *arXiv preprint arXiv:2412.15832*, 2024b.
- Roberto Molinaro. EPT-2.1 Helios: Europe’s most accurate solar model. Jua blog, <https://jua.ai/company/blog/ept2-helios>, July 2026. Published 15 July 2026; accessed 18 July 2026.
- Roberto Molinaro, Samuel Lanthaler, Bogdan Raonić, Tobias Rohner, Victor Armegioiu, Stephan Simonis, Dana Grund, Yannick Ramic, Zhong Yi Wan, Fei Sha, Siddhartha Mishra, and Leonardo Zepeda-Núñez. Generative AI for fast and accurate statistical computation of fluids. *arXiv preprint arXiv:2409.18359*, 2024.
- Roberto Molinaro, Niall Siegenheim, Niels Poulsen, Jordan Dane Daubinet, Henry Martin, Mark Frey, Kevin Thiart, Alexander Jakob Dautel, Andreas Schlueter, Alex Grigoryev, Bogdan Danciu, Nikoo Ekhtiari, Bas Steunebrink, Leonie Wagner, and Marvin Vincent Gabler. EPT-2 technical report. *arXiv preprint arXiv:2507.09703*, 2025.
- Roberto Molinaro, Niall Siegenheim, Henry Martin, Mark Frey, Niels Poulsen, Philipp Seitz, and Marvin Vincent Gabler. Universal diffusion-based probabilistic downscaling. *arXiv preprint arXiv:2602.11893*, 2026.
- Leonardo Olivetti and Gabriele Messori. Advances and prospects of deep learning for medium-range extreme weather forecasting. *Geoscientific Model Development*, 17(6):2347–2358, 2024a. doi: 10.5194/gmd-17-2347-2024.
- Leonardo Olivetti and Gabriele Messori. Do data-driven models beat numerical models in forecasting weather extremes? A comparison of IFS HRES, Pangu-Weather, and GraphCast. *Geoscientific Model Development*, 17(21):7915–7962, 2024b. doi: 10.5194/gmd-17-7915-2024.
- Jaideep Pathak, Yair Cohen, Piyush Garg, Peter Harrington, Noah Brenowitz, Dale Durran, Morteza Mardani, Arash Vahdat, Shaoming Xu, Karthik Kashinath, and Michael Pritchard. Kilometer-scale convection-allowing model emulation using generative diffusion modeling. *Science Advances*, 12(5):eadv0423, 2026. doi: 10.1126/sciadv.adv0423.
- S. E. Perkins and L. V. Alexander. On the measurement of heat waves. *Journal of Climate*, 26(13):4500–4517, 2013. doi: 10.1175/JCLI-D-12-00383.1.
- Ilan Price, Alvaro Sanchez-Gonzalez, Ferran Alet, Tom R. Andersson, Andrew El-Kadi, Dominic Masters, Timo Ewalds, Jacklynn Stott, Shakir Mohamed, Peter Battaglia, Remi Lam, and Matthew Willson. Probabilistic weather forecasting with machine learning. *Nature*, 637(8044):84–90, 2025. doi: 10.1038/s41586-024-08252-9.
- Andreas Schlueter and Leonie Wagner. juaAI/stationbench: v0.1.1. Zenodo, <https://doi.org/10.5281/zenodo.14671604>, January 2025. Software, MIT license; <https://github.com/juaAI/stationbench>.
- Niall Siegenheim. EPT2-HRRR: The first operational high-resolution AI weather model for Europe. Jua blog, <https://jua.ai/company/blog/ept2-hrrr>, October 2025. Published 20 October 2025; accessed 18 July 2026.
- Christopher Subich, Syed Zahid Husain, Leo Separovic, and Jing Yang. Fixing the double penalty in data-driven weather forecasting through a modified spherical harmonic loss function. *arXiv preprint arXiv:2501.19374*, 2025. Accepted at the International Conference on Machine Learning (ICML) 2025.

540 Xuebin Zhang, Lisa Alexander, Gabriele C. Hegerl, Philip Jones, Albert Klein Tank, Thomas C.
541 Peterson, Blair Trewin, and Francis W. Zwiers. Indices for monitoring changes in extremes based
542 on daily temperature and precipitation data. *WIREs Climate Change*, 2(6):851–870, 2011. doi:
543 10.1002/wcc.147.

544 Zhongwei Zhang, Erich Fischer, Jakob Zscheischler, and Sebastian Engelke. Physics-based mod-
545 els outperform AI weather forecasts of record-breaking extremes. *Science Advances*, 12(18):
546 eaec1433, 2026. doi: 10.1126/sciadv.aec1433.

547
548
549
550
551
552
553
554
555
556
557
558
559
560
561
562
563
564
565
566
567
568
569
570
571
572
573
574
575
576
577
578
579
580
581
582
583
584
585
586
587
588
589
590
591
592
593

A SUPPLEMENTARY MATERIAL

This supplement provides the model and regime definitions omitted from the main text, lead-time and short-range robustness checks for Track A, the matched regional comparison in Track B, complete numerical tables, and country-level regime profiles.

A.1 EVALUATION SETUP

Table S1 identifies the forecasting systems and their broad model classes. Tables S2 and S3 give the observed-value intervals and meteorological labels used throughout the evaluation.

Table S1: Forecast models included in the evaluation. ECMWF IFS is the reference model.

Model	Model class	Forecast type	Domain
ECMWF IFS (reference)	Physical	Deterministic	Global
ECMWF ENS	Physical	Ensemble	Global
NOAA GFS	Physical	Deterministic	Global
DWD ICON Global	Physical	Deterministic	Global
DWD ICON-EU	Physical	Deterministic	Regional
ECMWF AIFS	Regression AI	Deterministic	Global
Microsoft Aurora	Regression AI	Deterministic	Global
Jua EPT-2 Reasoning	Regression AI	Deterministic	Global
Jua EPT-2e	Regression AI	Ensemble	Global
Jua EPT-2.1 Helios	Regression AI (solar)	Deterministic	Regional
ECMWF AIFS ENS	Probabilistic AI	Ensemble	Global
Jua EPT-2 HRRR	Generative AI	Ensemble	Regional
Jua EPT-2.1 Europa	Generative AI	Ensemble	Regional

Table S2: Observed-value regimes for wind, temperature, and solar radiation. For solar radiation, the percentiles refer to positive daytime GHI.

Regime	Definition	Wind	Temperature	Solar
All	–	all conditions	all conditions	all daytime
Very low	$y < P5$	calm	severe cold	heavily overcast
Low	$P5 \leq y < P25$	light air	cold	overcast
Typical	$P25 \leq y < P75$	moderate	moderate	mixed sky
High	$P75 \leq y < P95$	fresh to strong	warm	mostly sunny
Very high	$y \geq P95$	gale force	heat	near clear sky

Table S3: Observed-value regimes for hourly precipitation. Wet-hour percentiles are conditional on ERA5 accumulations of at least 0.1 mm.

Regime	Definition	Interpretation
All	–	all hours
Dry	$y < 0.1$ mm	dry hour
Light	$0.1 \text{ mm} \leq y < P50$	light rain
Moderate	$P50 \leq y < P75$	moderate rain
High	$P75 \leq y < P95$	heavy rain
Heavy	$y \geq P95$	most intense hours

A.2 TRACK A LEAD-TIME, SHORT-RANGE, AND PRECIPITATION ROBUSTNESS

Figures S1–S4 show skill as a function of forecast lead for all conditions and the relevant upper or lower tail. Figure S5 repeats the regime comparison over 1–12 h, and Table S4 reports heat-regime skill by country. Figures 2–S7 and Tables S5 and S12 provide the full-horizon, short-range, and categorical precipitation checks.

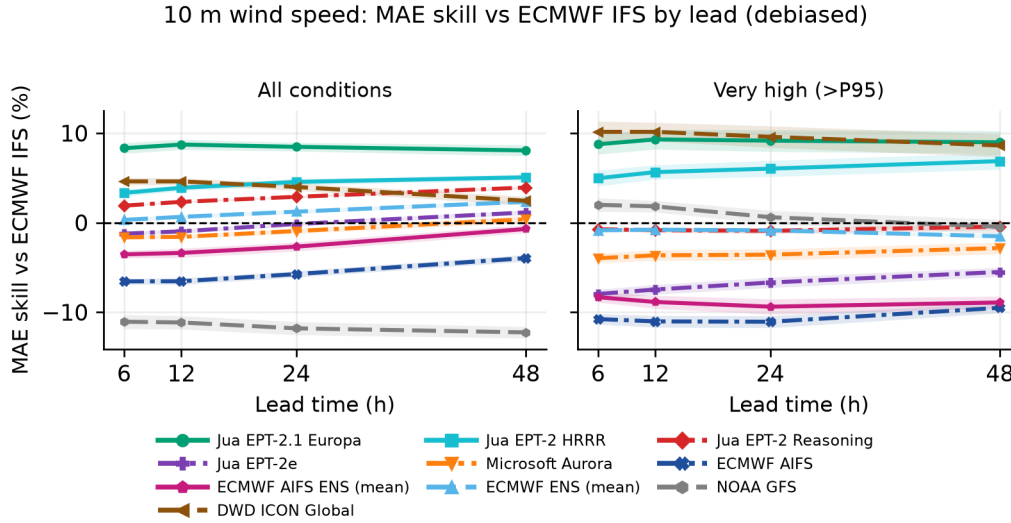


Figure S1: MAE skill relative to ECMWF IFS by lead time for 10 m wind. The left panel shows all conditions; the right panel shows the gale-force regime, $y \geq P95$. Shading denotes ± 1 jackknife SE.

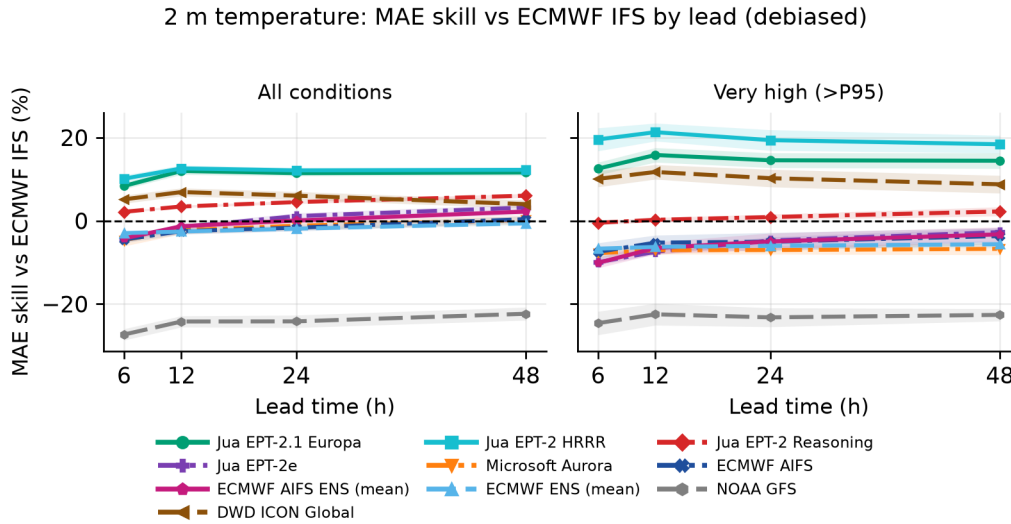


Figure S2: As Figure S1, for 2 m temperature. The right panel shows the heat regime, $y \geq P95$.

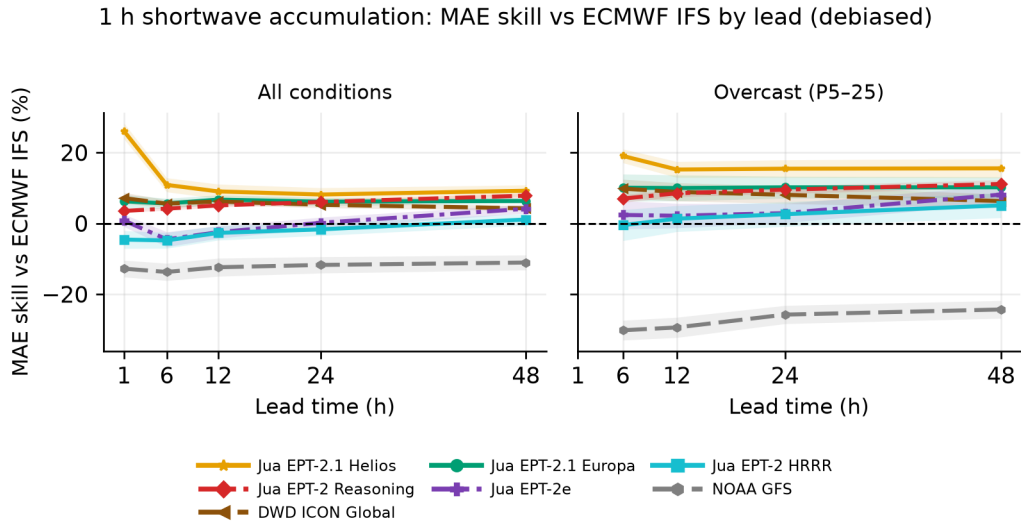


Figure S3: Track A MAE skill relative to ECMWF IFS by lead time for hourly solar radiation. The left panel shows all daytime conditions; the right panel shows the overcast regime, $P5 \leq y < P25$. AIFS, AIFS ENS, Aurora, and ECMWF ENS are omitted because they lack compatible hourly solar output. Shading denotes ± 1 jackknife SE.

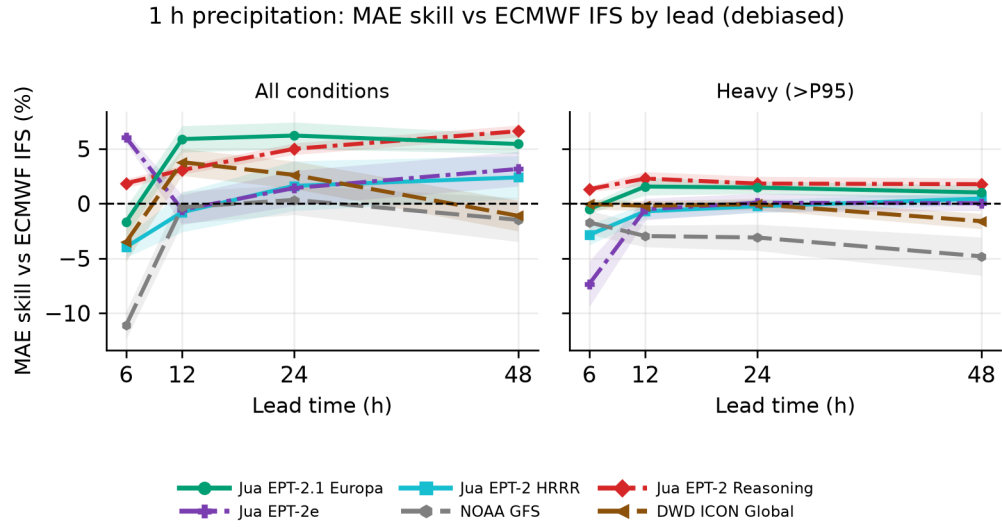


Figure S4: Track A MAE skill relative to ECMWF IFS by lead time for hourly precipitation. The left panel shows all conditions; the right panel shows the heavy regime, $y \geq P95$. Shading denotes ± 1 jackknife SE.

Table S4: Track A MAE skill (%) by country in the heat regime ($y \geq P95$), aggregated over 6–48 h after bias correction. NOAA GFS is negative in every country with available observations.

Model	DE	FR	GB	ES	IT	PL	NL	BE	CH	NO	AT	CZ	DK
Jua EPT-2.1 Europa	+6.6	+16.1	-1.2	+6.4	+17.2	+6.2	-3.4	-3.0	+31.1	+5.5	+28.2	+16.6	-4.4
Jua EPT-2 HRRR	+10.5	+21.5	+4.0	+10.5	+27.6	+5.6	+1.8	-0.6	+29.4	+14.2	+16.6	+17.7	+12.2
Jua EPT-2 Reasoning	+6.6	+2.6	+4.3	-2.6	+0.1	+3.9	+3.8	-1.0	-1.3	-5.2	-3.1	+7.5	+2.7
Jua EPT-2e	+6.1	-2.2	-0.3	-15.1	-10.4	-2.4	+0.6	-2.8	-2.5	-12.2	-16.3	-1.3	+1.7
Microsoft Aurora	+2.1	-4.5	-7.5	-18.7	-7.8	-0.9	+0.7	-11.8	-3.3	-18.0	-15.4	-7.0	+1.8
ECMWF AIFS	+0.8	-1.6	-10.2	-12.7	-14.6	+1.9	-0.4	+6.3	+1.9	-3.4	-25.0	-3.1	+10.3
ECMWF AIFS ENS (mean)	+5.3	-3.1	-6.7	-10.6	-12.6	-3.1	-3.9	+5.1	+4.6	-3.2	-38.7	+2.1	+4.4
ECMWF ENS (mean)	-2.1	-4.6	-5.1	-12.1	-4.0	-8.6	+3.4	-2.9	-10.7	-9.8	-10.6	-1.5	-9.2
NOAA GFS	-18.4	-23.6	-37.0	-25.1	-13.2	-24.3	-21.9	-12.2	-10.2	-43.1	-25.3	-31.4	-24.7
DWD ICON Global	+6.9	+8.8	-4.5	+3.6	+12.8	+4.8	-2.7	+10.0	+22.7	+18.1	+16.8	+15.5	-6.5
DWD ICON-EU†	+13.1	+18.4	+1.3	+7.4	+20.1	+9.1	-2.6	+14.1	+34.6	+6.6	+28.1	+17.8	-0.7

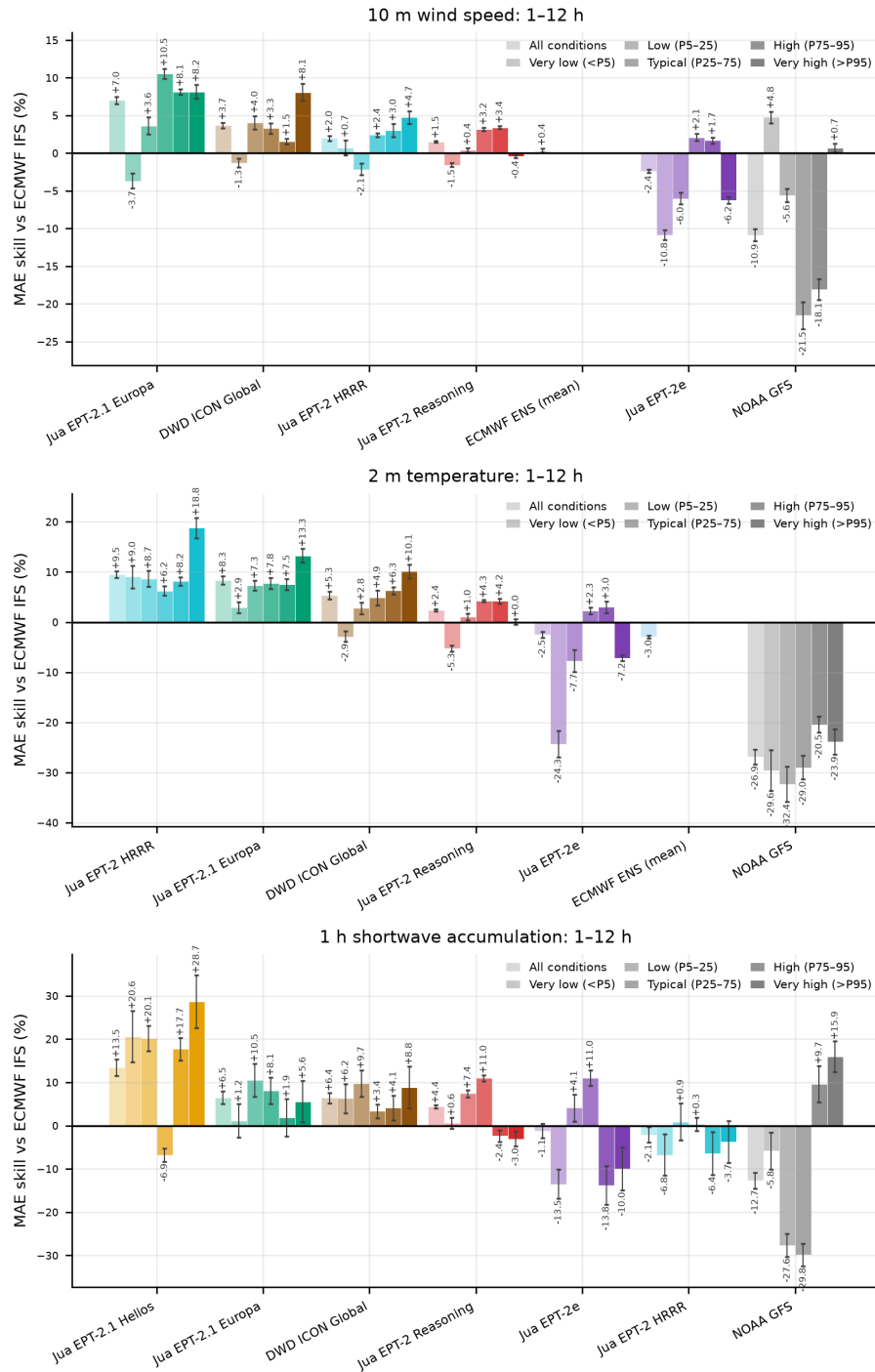


Figure S5: Track A MAE skill relative to ECMWF IFS by observed-value regime over 1–12 h, with ± 1 jack-knife SE. Wind and temperature are restricted to models with hourly output; solar radiation includes EPT-2.1 Helios.

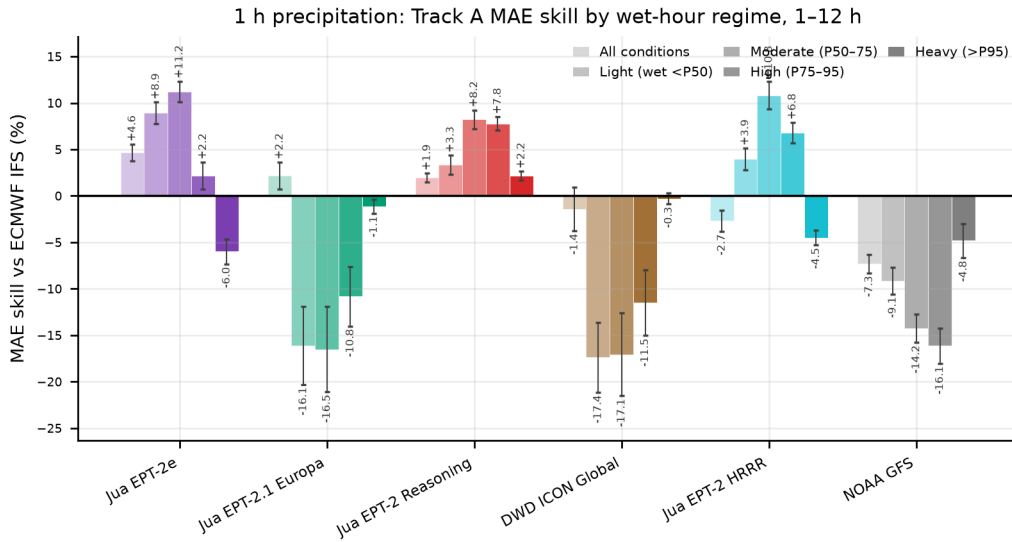


Figure S6: As Figure 2, restricted to the 1–12 h forecast range.

Table S5: Track A MAE skill (%) relative to ECMWF IFS for hourly precipitation by observed-value regime, aggregated over 1–48 h after multiplicative bias correction. Parentheses give the leave-one-month-out jackknife SE.

Model	All	Dry	<P50	P50–75	P75–95	>P95
Jua EPT-2 Reasoning	+5.1 (±0.4)	+2.2 (±1.3)	+9.3 (±0.6)	+14.0 (±0.9)	+10.6 (±0.8)	+1.7 (±0.5)
Jua EPT-2 HRRR	+1.3 (±1.3)	-6.1 (±3.3)	+9.3 (±1.8)	+15.2 (±2.2)	+10.2 (±1.2)	-1.5 (±0.5)
Jua EPT-2e	+3.3 (±1.1)	+1.5 (±4.8)	+7.7 (±1.1)	+14.8 (±1.7)	+8.8 (±1.5)	-1.9 (±0.9)
Jua EPT-2.1 Europa	+4.2 (±0.7)	+21.0 (±2.7)	-11.1 (±3.0)	-10.5 (±3.4)	-5.9 (±2.5)	+0.2 (±0.6)
NOAA GFS	-4.3 (±1.0)	-0.2 (±1.8)	-5.1 (±2.0)	-9.3 (±1.8)	-11.2 (±1.4)	-3.7 (±1.2)
DWD ICON Global	-1.4 (±1.7)	+8.3 (±2.7)	-14.4 (±2.7)	-14.4 (±3.0)	-9.8 (±2.2)	+0.0 (±0.4)

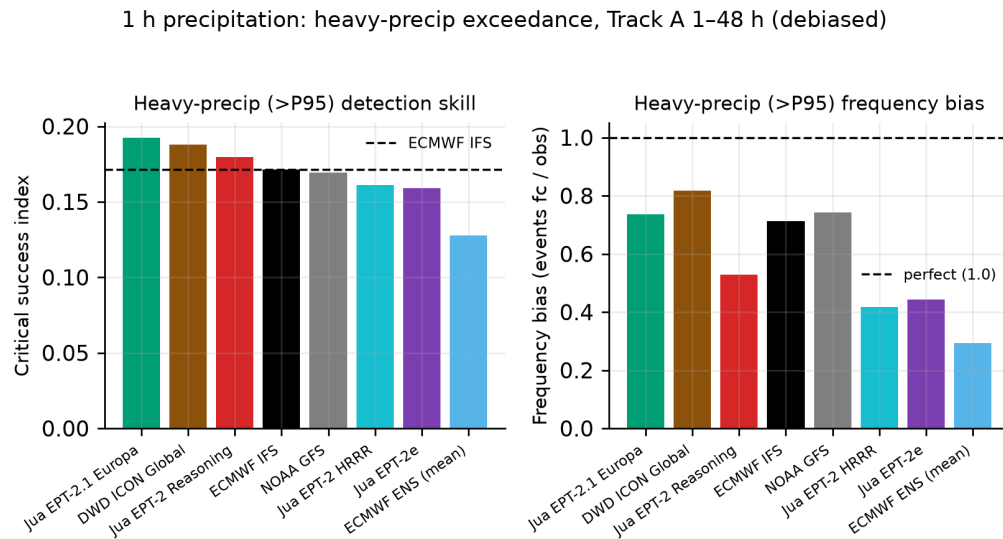


Figure S7: Detection of heavy precipitation ($y \geq P95$) on Track A over 1–48 h. Left: critical success index, with ECMWF IFS shown by the dashed line. Right: frequency bias, with unity denoting the correct event frequency. All models underpredict the frequency of heavy precipitation.

A.3 MATCHED REGIONAL COMPARISON

Track B robustness beyond the full-horizon comparison in the main text is shown in Figure S8 for 1–12 h and in Figure S9 as a function of lead. Figures S10 and S11 give the corresponding precipitation results.

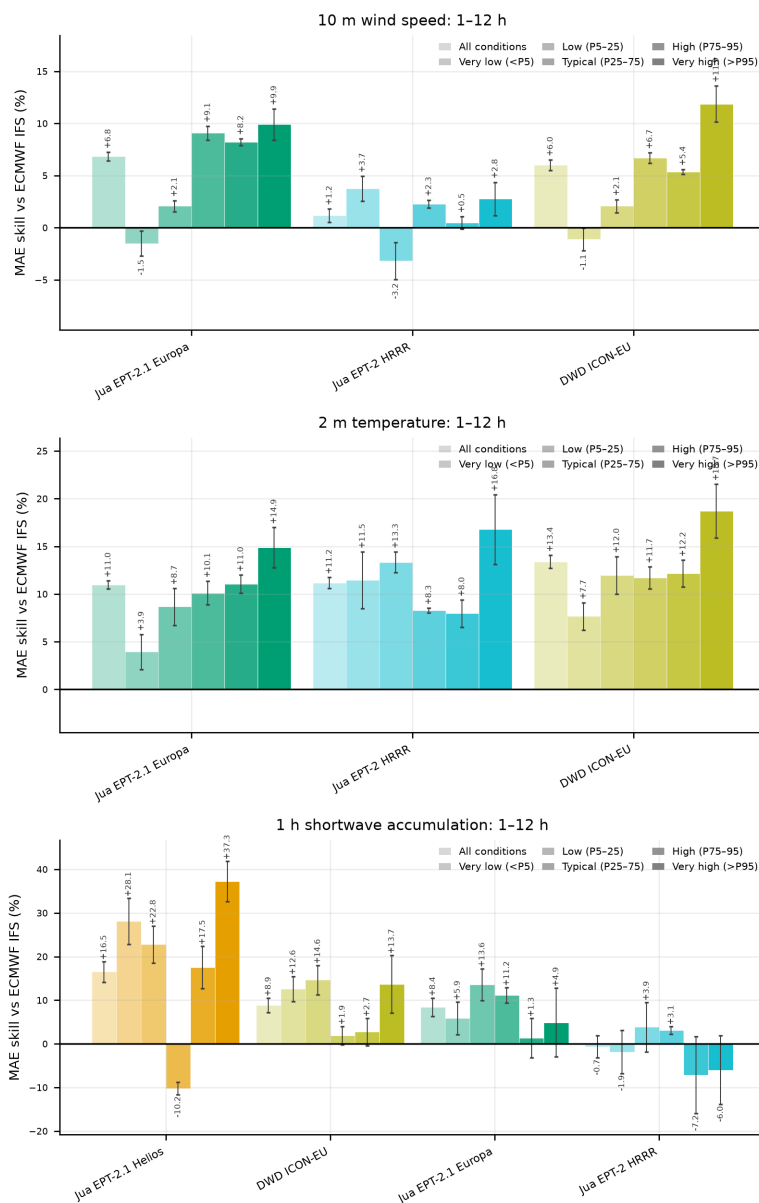


Figure S8: As Figure 4, restricted to the 1–12 h forecast range.

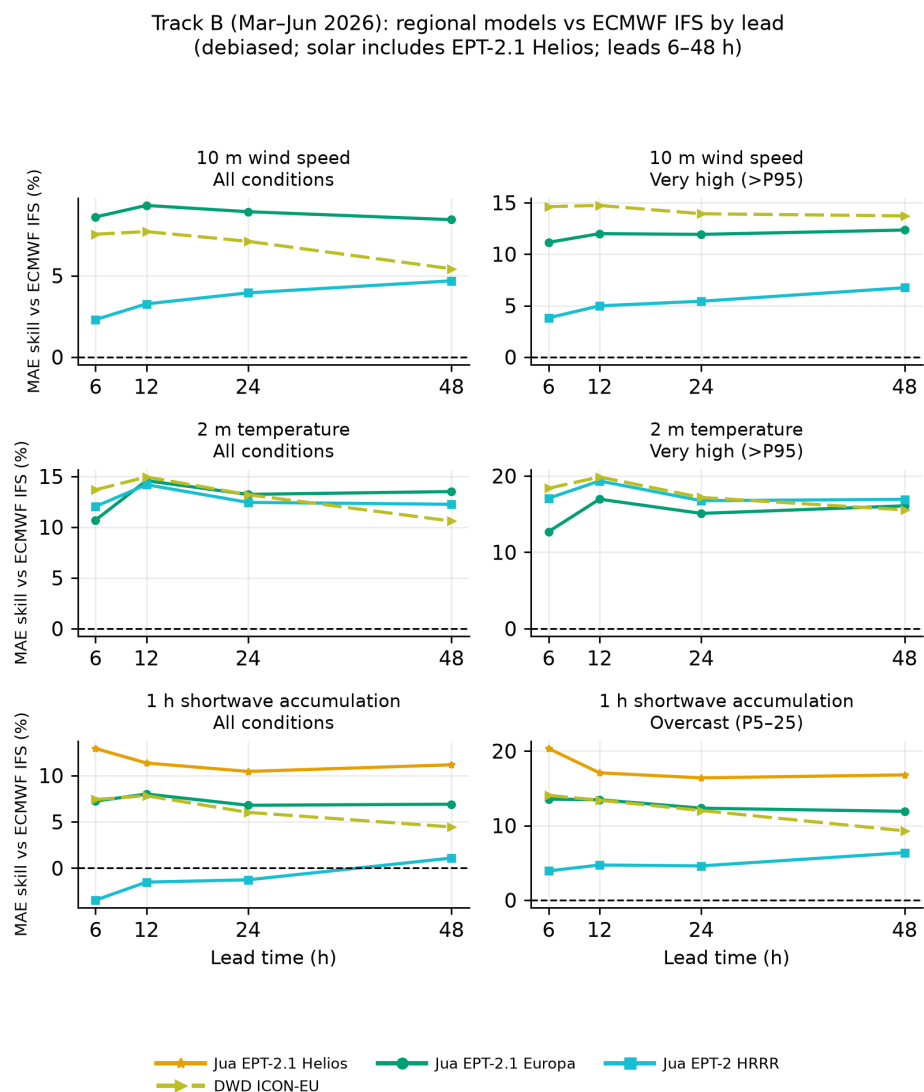


Figure S9: Track B MAE skill relative to ECMWF IFS by lead time. Rows show 10 m wind, 2 m temperature, and hourly solar radiation. Columns compare all conditions with gale-force wind, heat, and overcast conditions, respectively. Solar radiation additionally includes EPT-2.1 Helios. Plotted leads are 6, 12, 24, and 48 h; shading denotes ± 1 jackknife SE.

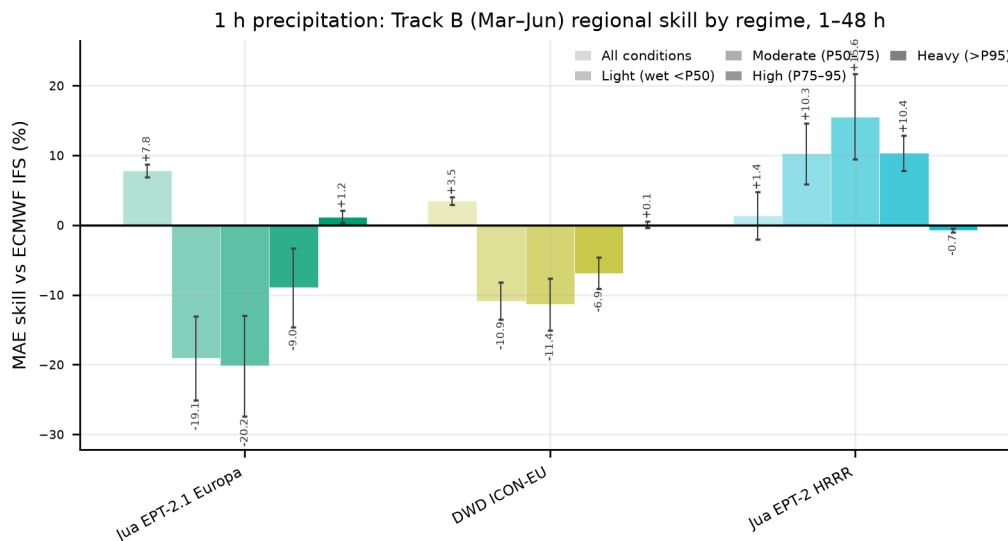


Figure S10: Track B MAE skill relative to ECMWF IFS for hourly precipitation by observed-value regime, aggregated over 1–48 h from March to June 2026, with ± 1 jackknife SE.

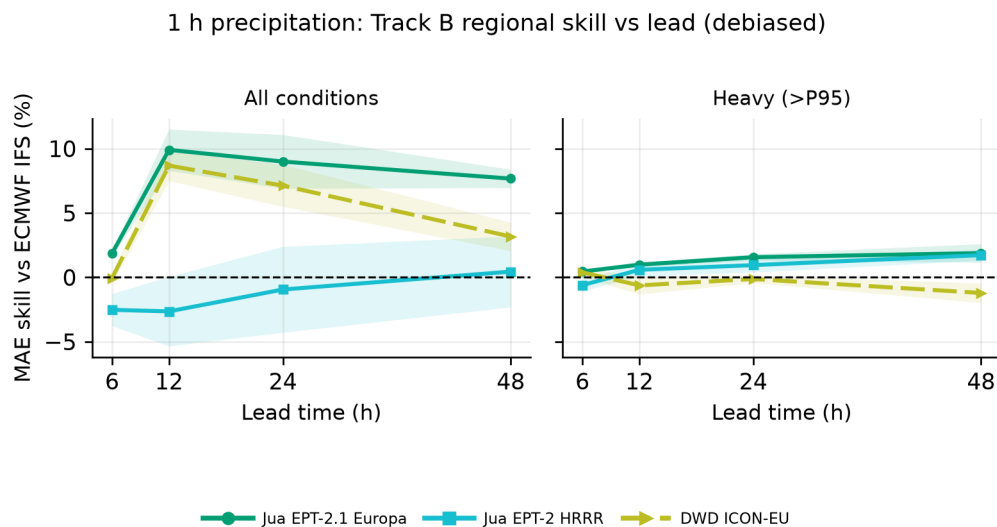


Figure S11: Track B precipitation skill relative to ECMWF IFS by lead time. The left panel shows all conditions; the right panel shows the heavy regime, $y \geq P95$. Shading denotes ± 1 jackknife SE.

A.4 COMPLETE NUMERICAL RESULTS

Tables S6–S15 report Track A regime, short-range, and lead-time values. Tables S16–S22 report the corresponding Track B values; Tables S23 and S24 complete the regional precipitation and heavy-event detection results.

Table S6: Track A MAE skill (%), 6–48 h, by regime – Figure 1, wind. Parentheses: jackknife SE.

Model	All	<P5	P5–25	P25–75	P75–95	>P95
Jua EPT-2.1 Europa	+8.4 (0.5)	-1.7 (1.4)	+6.4 (0.9)	+11.3 (0.7)	+8.5 (0.4)	+9.0 (1.2)
Jua EPT-2 HRRR	+4.5 (0.3)	+2.7 (1.0)	+1.4 (0.9)	+5.1 (0.3)	+4.9 (1.0)	+6.1 (0.9)
Jua EPT-2 Reasoning	+3.0 (0.1)	-0.7 (0.4)	+2.8 (0.5)	+5.6 (0.3)	+4.5 (0.3)	-0.7 (0.3)
Jua EPT-2e	-0.0 (0.2)	-9.1 (1.0)	-1.4 (1.1)	+5.6 (0.5)	+2.9 (0.6)	-6.6 (0.5)
Microsoft Aurora	-0.7 (0.3)	-6.6 (1.1)	-3.9 (1.1)	+1.9 (0.8)	+2.6 (0.8)	-3.4 (0.6)
ECMWF AIFS	-5.4 (0.4)	-9.8 (1.2)	-8.0 (1.3)	-2.1 (0.6)	-1.4 (0.8)	-10.6 (0.6)
ECMWF AIFS ENS (mean)	-2.3 (0.5)	-12.0 (1.2)	-3.3 (1.5)	+3.8 (0.7)	-0.1 (0.7)	-9.0 (0.7)
ECMWF ENS (mean)	+1.4 (0.2)	-0.9 (0.7)	+1.5 (0.6)	+3.1 (0.4)	+2.1 (0.4)	-1.0 (0.2)
NOAA GFS	-11.7 (0.7)	+1.8 (1.4)	-9.5 (1.4)	-21.8 (1.4)	-16.3 (1.3)	+0.6 (0.7)
DWD ICON Global	+3.6 (0.4)	-4.1 (1.2)	+2.8 (1.0)	+2.6 (0.7)	+2.5 (0.7)	+9.5 (1.1)

Table S7: Track A MAE skill (%), 6–48 h – Figure 1, temperature.

Model	All	<P5	P5–25	P25–75	P75–95	>P95
Jua EPT-2.1 Europa	+11.5 (0.7)	+4.7 (1.0)	+10.8 (1.3)	+11.0 (1.0)	+11.8 (1.1)	+15.0 (1.7)
Jua EPT-2 HRRR	+12.1 (0.4)	+9.5 (2.4)	+11.5 (1.2)	+9.4 (0.9)	+11.4 (1.0)	+19.6 (2.2)
Jua EPT-2 Reasoning	+4.6 (0.4)	-6.9 (1.3)	+2.6 (1.3)	+7.4 (0.4)	+7.0 (0.8)	+1.1 (0.7)
Jua EPT-2e	+0.8 (0.7)	-24.3 (3.6)	-5.0 (2.8)	+6.1 (0.8)	+7.4 (1.1)	-5.0 (0.7)
Microsoft Aurora	-1.3 (1.1)	-22.2 (4.4)	-7.0 (4.6)	+4.1 (1.0)	+3.8 (1.6)	-6.9 (1.0)
ECMWF AIFS	-1.4 (0.6)	-27.3 (3.6)	-5.1 (2.3)	+2.3 (1.0)	+4.9 (1.4)	-4.9 (2.0)
ECMWF AIFS ENS (mean)	+0.2 (0.6)	-24.9 (3.2)	-3.1 (2.0)	+4.5 (0.9)	+6.2 (0.9)	-5.1 (1.7)
ECMWF ENS (mean)	-1.7 (0.3)	-12.2 (1.0)	-3.3 (0.9)	+1.4 (0.4)	+0.3 (0.8)	-6.1 (0.5)
NOAA GFS	-23.9 (1.4)	-25.3 (3.5)	-26.3 (2.9)	-26.5 (2.5)	-17.8 (2.1)	-22.8 (2.0)
DWD ICON Global	+5.5 (0.9)	-3.8 (1.2)	+3.0 (1.7)	+4.1 (1.5)	+8.3 (0.9)	+10.2 (2.0)

Table S8: Track A MAE skill (%), 1–48 h hourly – Figure 1, solar. Parentheses: jackknife SE.

Model	All	<P5	P5–25	P25–75	P75–95	>P95
Jua EPT-2.1 Helios	+10.2 (1.7)	+15.4 (5.0)	+16.4 (3.4)	-8.2 (1.1)	+13.7 (2.9)	+24.8 (5.4)
Jua EPT-2.1 Europa	+7.1 (1.0)	+3.1 (4.0)	+11.0 (3.1)	+9.2 (2.3)	+1.9 (4.3)	+4.4 (4.8)
Jua EPT-2 HRRR	+0.6 (1.4)	-3.6 (4.6)	+3.8 (3.7)	+4.5 (1.0)	-4.6 (4.8)	-4.1 (4.9)
Jua EPT-2 Reasoning	+6.3 (0.5)	+0.9 (1.3)	+9.5 (0.8)	+14.0 (0.7)	-0.8 (1.9)	-2.6 (2.6)
Jua EPT-2e	+2.0 (1.4)	-10.6 (3.3)	+5.9 (2.3)	+13.5 (1.6)	-7.5 (4.1)	-7.6 (6.0)
NOAA GFS	-11.2 (1.6)	-3.3 (4.4)	-24.3 (2.1)	-28.3 (2.5)	+8.8 (3.4)	+16.7 (4.2)
DWD ICON Global	+5.6 (0.8)	+6.5 (3.3)	+8.1 (2.6)	+3.1 (1.5)	+4.2 (3.3)	+6.9 (4.2)

Table S9: Track A MAE skill (%), 1–12 h – Figure S5, wind.

Model	All	<P5	P5–25	P25–75	P75–95	>P95
Jua EPT-2.1 Europa	+7.0 (0.5)	-3.7 (1.0)	+3.6 (1.1)	+10.5 (0.7)	+8.1 (0.3)	+8.2 (0.9)
Jua EPT-2 HRRR	+2.0 (0.3)	+0.7 (1.0)	-2.1 (0.8)	+2.4 (0.2)	+3.0 (0.9)	+4.7 (0.9)
Jua EPT-2 Reasoning	+1.5 (0.1)	-1.5 (0.2)	+0.4 (0.3)	+3.2 (0.2)	+3.4 (0.2)	-0.4 (0.2)
Jua EPT-2e	-2.4 (0.2)	-10.8 (0.7)	-6.0 (0.7)	+2.1 (0.5)	+1.7 (0.4)	-6.2 (0.4)
ECMWF ENS (mean)	+0.4 (0.3)	–	–	–	–	–
NOAA GFS	-10.9 (0.8)	+4.8 (0.8)	-5.6 (0.9)	-21.5 (1.8)	-18.1 (1.4)	+0.7 (0.6)
DWD ICON Global	+3.7 (0.3)	-1.3 (0.6)	+4.0 (0.9)	+3.3 (0.7)	+1.5 (0.4)	+8.1 (1.1)

Table S10: Track A MAE skill (%), 1–12 h – Figure S5, temperature.

Model	All	<P5	P5–25	P25–75	P75–95	>P95
Jua EPT-2.1 Europa	+8.3 (0.8)	+2.9 (1.1)	+7.3 (1.0)	+7.8 (1.1)	+7.5 (1.1)	+13.3 (1.4)
Jua EPT-2 HRRR	+9.5 (0.6)	+9.0 (2.3)	+8.7 (1.6)	+6.2 (1.0)	+8.2 (0.8)	+18.8 (2.0)
Jua EPT-2 Reasoning	+2.4 (0.2)	-5.3 (0.6)	+1.0 (0.6)	+4.3 (0.2)	+4.2 (0.5)	+0.0 (0.5)
Jua EPT-2e	-2.5 (0.6)	-24.3 (2.7)	-7.7 (2.2)	+2.3 (0.7)	+3.0 (1.1)	-7.2 (0.5)
ECMWF ENS (mean)	-3.0 (0.2)	–	–	–	–	–
NOAA GFS	-26.9 (1.5)	-29.6 (4.0)	-32.4 (3.5)	-29.0 (2.4)	-20.5 (1.6)	-23.9 (2.5)
DWD ICON Global	+5.3 (0.8)	-2.9 (1.0)	+2.8 (1.2)	+4.9 (1.5)	+6.3 (0.7)	+10.1 (1.4)

Table S11: Track A MAE skill (%), 1–12 h – Figure S5, solar.

Model	All	<P5	P5–25	P25–75	P75–95	>P95
Jua EPT-2.1 Helios	+13.5 (1.9)	+20.6 (5.9)	+20.1 (2.9)	-6.9 (1.5)	+17.7 (2.6)	+28.7 (6.1)
Jua EPT-2.1 Europa	+6.5 (1.4)	+1.2 (3.9)	+10.5 (3.8)	+8.1 (3.0)	+1.9 (4.3)	+5.6 (4.8)
Jua EPT-2 HRRR	-2.1 (1.8)	-6.8 (4.7)	+0.9 (4.3)	+0.3 (1.5)	-6.4 (5.0)	-3.7 (4.8)
Jua EPT-2 Reasoning	+4.4 (0.4)	+0.6 (1.3)	+7.4 (0.8)	+11.0 (0.7)	-2.4 (1.3)	-3.0 (1.7)
Jua EPT-2e	-1.1 (1.7)	-13.5 (3.4)	+4.1 (3.1)	+11.0 (1.7)	-13.8 (4.5)	-10.0 (5.0)
NOAA GFS	-12.7 (1.9)	-5.8 (4.2)	-27.6 (2.7)	-29.8 (2.6)	+9.7 (4.2)	+15.9 (3.6)
DWD ICON Global	+6.4 (1.2)	+6.2 (3.4)	+9.7 (3.0)	+3.4 (1.6)	+4.1 (2.9)	+8.8 (4.8)

Table S12: Track A precipitation MAE skill (%), 1–12 h (Figure S6). Parentheses: jackknife SE.

Model	All	Dry	<P50	P50–75	P75–95	>P95
Jua EPT-2 Reasoning	+1.9 (± 0.5)	-2.9 (± 1.0)	+3.3 (± 1.0)	+8.2 (± 1.0)	+7.8 (± 0.7)	+2.2 (± 0.5)
Jua EPT-2 HRRR	-2.7 (± 1.1)	-10.9 (± 3.8)	+3.9 (± 1.2)	+10.8 (± 1.5)	+6.8 (± 1.1)	-4.5 (± 0.8)
Jua EPT-2e	+4.6 (± 0.9)	+12.4 (± 3.4)	+8.9 (± 1.1)	+11.2 (± 1.1)	+2.2 (± 1.5)	-6.0 (± 1.3)
Jua EPT-2.1 Europa	+2.2 (± 1.5)	+22.0 (± 3.2)	-16.1 (± 4.2)	-16.5 (± 4.6)	-10.8 (± 3.2)	-1.1 (± 0.7)
NOAA GFS	-7.3 (± 1.0)	-3.4 (± 3.6)	-9.1 (± 1.4)	-14.2 (± 1.5)	-16.1 (± 1.9)	-4.8 (± 1.8)
DWD ICON Global	-1.4 (± 2.3)	+11.4 (± 3.3)	-17.4 (± 3.8)	-17.1 (± 4.5)	-11.5 (± 3.5)	-0.3 (± 0.6)

Table S13: Track A MAE skill (%) by lead – Figure S1, wind. Point estimates (no SE in per-lead tables).

Model	6 h		12 h		24 h		48 h	
	All	>P95	All	>P95	All	>P95	All	>P95
EPT-2.1 Europa	+8.3	+8.8	+8.7	+9.3	+8.5	+9.2	+8.1	+9.0
EPT-2 HRRR	+3.3	+5.0	+3.9	+5.6	+4.6	+6.0	+5.1	+6.9
EPT-2 Reasoning	+1.9	-0.7	+2.3	-0.8	+2.9	-0.9	+3.9	-0.4
EPT-2e	-1.2	-8.0	-1.0	-7.5	-0.1	-6.7	+1.1	-5.5
Aurora	-1.6	-4.0	-1.6	-3.6	-0.9	-3.6	+0.4	-2.8
AIFS	-6.6	-10.8	-6.5	-11.0	-5.7	-11.1	-4.0	-9.5
AIFS ENS	-3.5	-8.3	-3.4	-8.9	-2.7	-9.4	-0.7	-8.9
ENS	+0.3	-0.9	+0.7	-0.8	+1.2	-0.9	+2.3	-1.5
GFS	-11.1	+2.0	-11.1	+1.8	-11.8	+0.6	-12.3	-0.5
ICON Global	+4.6	+10.1	+4.6	+10.1	+4.0	+9.6	+2.4	+8.6

Table S14: Track A MAE skill (%) by lead – Figure S2, temperature. Point estimates (no SE in per-lead tables).

Model	6 h		12 h		24 h		48 h	
	All	>P95	All	>P95	All	>P95	All	>P95
EPT-2.1 Europa	+8.4	+12.5	+12.0	+15.8	+11.5	+14.5	+11.6	+14.4
EPT-2 HRRR	+10.1	+19.5	+12.6	+21.3	+12.1	+19.4	+12.2	+18.4
EPT-2 Reasoning	+2.2	-0.5	+3.4	+0.2	+4.5	+0.9	+6.0	+2.2
EPT-2e	-4.2	-10.0	-1.5	-7.5	+1.1	-4.7	+3.2	-2.7
Aurora	-4.6	-7.8	-2.3	-7.1	-1.2	-7.0	+0.2	-6.7
AIFS	-4.5	-7.5	-2.5	-5.2	-1.7	-5.0	+0.4	-3.7
AIFS ENS	-4.3	-10.1	-1.3	-6.6	+0.1	-5.0	+2.2	-3.3
ENS	-3.0	-6.7	-2.6	-6.3	-1.9	-6.0	-0.6	-5.6
GFS	-27.3	-24.6	-24.2	-22.5	-24.2	-23.2	-22.3	-22.6
ICON Global	+5.1	+10.1	+6.9	+11.7	+6.0	+10.2	+4.0	+8.8

Table S15: Track A MAE skill (%) by lead – Figure S3, solar (Ovc. = overcast, P5–P25). Point estimates; SEs in regime tables.

Model	1 h		6 h		12 h		24 h		48 h	
	All	Ovc.	All	Ovc.	All	Ovc.	All	Ovc.	All	Ovc.
EPT-2.1 Helios	+26.1	–	+10.9	+19.0	+9.0	+15.2	+8.1	+15.4	+9.2	+15.5
EPT-2.1 Europa	+6.2	–	+5.6	+10.1	+6.7	+10.0	+6.1	+10.2	+6.3	+10.2
ICON Global	+7.1	–	+5.6	+9.8	+6.2	+8.9	+5.2	+8.1	+4.3	+6.3
EPT-2 Reasoning	+3.5	–	+4.2	+7.0	+5.1	+8.5	+6.0	+9.5	+7.8	+11.1
EPT-2e	+0.8	–	-4.5	+2.4	-2.5	+2.2	+0.2	+2.9	+4.1	+8.1
EPT-2 HRRR	-4.6	–	-4.8	-0.5	-2.7	+1.3	-1.7	+2.6	+1.1	+5.1
GFS	-12.8	–	-13.7	-30.2	-12.4	-29.4	-11.7	-25.8	-11.1	-24.3

Table S16: Track B wind MAE skill (%), 1–48 h, by regime (Figure 4).

Model	All	<P5	P5–25	P25–75	P75–95	>P95
Jua EPT-2.1 Europa	+7.2 (0.4)	+0.1 (0.8)	+3.0 (0.3)	+8.9 (0.6)	+8.2 (0.3)	+10.5 (1.4)
Jua EPT-2 HRRR	+2.4 (0.6)	+4.9 (1.0)	-1.8 (1.7)	+3.5 (0.4)	+1.8 (0.4)	+4.1 (1.5)
DWD ICON-EU	+5.3 (0.5)	-0.7 (1.0)	+1.8 (0.6)	+5.6 (0.5)	+3.9 (0.2)	+11.5 (1.6)

Table S17: Track B MAE skill (%), 1–48 h – Figure 4, temperature.

Model	All	<P5	P5–25	P25–75	P75–95	>P95
Jua EPT-2.1 Europa	+13.2 (0.6)	+5.9 (1.8)	+11.7 (2.0)	+12.3 (1.3)	+13.5 (0.6)	+16.6 (2.3)
Jua EPT-2 HRRR	+12.0 (0.7)	+12.8 (3.3)	+14.9 (1.4)	+9.2 (0.4)	+8.7 (1.3)	+17.2 (3.5)
DWD ICON-EU	+12.4 (0.8)	+7.7 (1.1)	+11.8 (2.3)	+10.6 (1.1)	+11.3 (1.5)	+17.2 (3.2)

Table S18: Track B MAE skill (%), 1–48 h – Figure 4, solar.

Model	All	<P5	P5–25	P25–75	P75–95	>P95
Jua EPT-2.1 Helios	+12.6 (2.5)	+21.6 (4.6)	+18.1 (5.5)	-11.3 (0.5)	+13.5 (5.8)	+32.2 (4.9)
Jua EPT-2.1 Europa	+7.8 (2.0)	+7.4 (3.9)	+12.8 (3.1)	+10.7 (1.7)	+0.1 (4.5)	+3.4 (7.8)
Jua EPT-2 HRRR	+0.8 (2.2)	-0.2 (5.1)	+4.8 (5.7)	+6.5 (0.3)	-6.1 (8.2)	-6.4 (7.7)
DWD ICON-EU	+7.2 (1.4)	+12.4 (3.3)	+12.4 (2.5)	+1.0 (2.2)	+0.9 (3.3)	+10.5 (5.7)

Table S19: Track B MAE skill (%) by lead – Figure S9, wind.

Model	6 h		12 h		24 h		48 h	
	All	>P95	All	>P95	All	>P95	All	>P95
EPT-2.1 Europa	+8.6	+11.1	+9.3	+12.0	+8.9	+11.9	+8.4	+12.3
EPT-2 HRRR	+2.3	+3.8	+3.3	+5.0	+3.9	+5.4	+4.7	+6.7
ICON-EU	+7.5	+14.6	+7.7	+14.7	+7.1	+13.9	+5.4	+13.7

Table S20: Track B MAE skill (%) by lead – Figure S9, temperature.

Model	6 h		12 h		24 h		48 h	
	All	>P95	All	>P95	All	>P95	All	>P95
EPT-2.1 Europa	+10.7	+12.7	+14.6	+17.0	+13.2	+15.1	+13.5	+16.1
EPT-2 HRRR	+12.0	+17.1	+14.2	+19.3	+12.4	+16.8	+12.2	+16.9
ICON-EU	+13.7	+18.4	+14.9	+19.9	+13.2	+17.2	+10.6	+15.5

Table S21: Track B MAE skill (%) by lead – Figure S9, solar (Ovc. = overcast, P5–P25).

Model	6 h		12 h		24 h		48 h	
	All	Ovc.	All	Ovc.	All	Ovc.	All	Ovc.
EPT-2.1 Helios	+12.9	+20.3	+11.4	+17.1	+10.4	+16.4	+11.2	+16.8
EPT-2.1 Europa	+7.2	+13.6	+8.0	+13.5	+6.8	+12.3	+6.9	+11.9
EPT-2 HRRR	-3.5	+3.9	-1.5	+4.7	-1.3	+4.6	+1.1	+6.4
ICON-EU	+7.4	+14.1	+7.8	+13.4	+6.0	+12.0	+4.4	+9.3

Table S22: Track B precipitation MAE skill (%) by lead (Figure S11). Point estimates.

Model	6 h		12 h		24 h		48 h	
	All	>P95	All	>P95	All	>P95	All	>P95
Jua EPT-2.1 Europa	+1.9	+0.5	+9.9	+1.0	+9.0	+1.6	+7.7	+1.9
Jua EPT-2 HRRR	-2.5	-0.6	-2.6	+0.6	-0.9	+1.0	+0.4	+1.7
DWD ICON-EU	-0.1	+0.4	+8.7	-0.6	+7.1	-0.1	+3.2	-1.2

Table S23: Track B MAE skill (%) for 1 h precipitation by wet-hour regime, 1 to 48 h, debiased – Figure S10. Parentheses: jackknife SE.

Model	All	Dry	<P50	P50–75	P75–95	>P95
Jua EPT-2.1 Europa	+7.8 (± 0.9)	+29.3 (± 1.4)	-19.1 (± 6.0)	-20.2 (± 7.2)	-9.0 (± 5.7)	+1.2 (± 0.8)
Jua EPT-2 HRRR	+1.4 (± 3.4)	-3.6 (± 9.7)	+10.3 (± 4.4)	+15.6 (± 6.1)	+10.4 (± 2.5)	-0.7 (± 0.3)
DWD ICON-EU	+3.5 (± 0.5)	+15.3 (± 0.3)	-10.9 (± 2.7)	-11.4 (± 3.8)	-6.9 (± 2.2)	+0.1 (± 0.5)

Table S24: Heavy-precipitation (>P95) detection on Track A, 1 to 48 h, debiased – Figure S7. POD: probability of detection; FAR: false-alarm ratio; CSI: critical success index; frequency bias is forecast over observed event count.

Model	POD	FAR	CSI	Freq. bias
Jua EPT-2.1 Europa	0.28	0.62	0.19	0.74
DWD ICON Global	0.29	0.65	0.19	0.82
Jua EPT-2 Reasoning	0.23	0.56	0.18	0.53
ECMWF IFS	0.25	0.65	0.17	0.71
NOAA GFS	0.25	0.66	0.17	0.74
Jua EPT-2 HRRR	0.20	0.53	0.16	0.42
Jua EPT-2e	0.20	0.55	0.16	0.44
ECMWF ENS (mean)	0.15	0.50	0.13	0.29

A.5 COUNTRY-LEVEL REGIME PROFILES

Figures S12–S15 show how the March–June regime-skill patterns vary across countries for the mixed regional/global comparison set. Countries without sufficient matched observations for a variable are omitted.

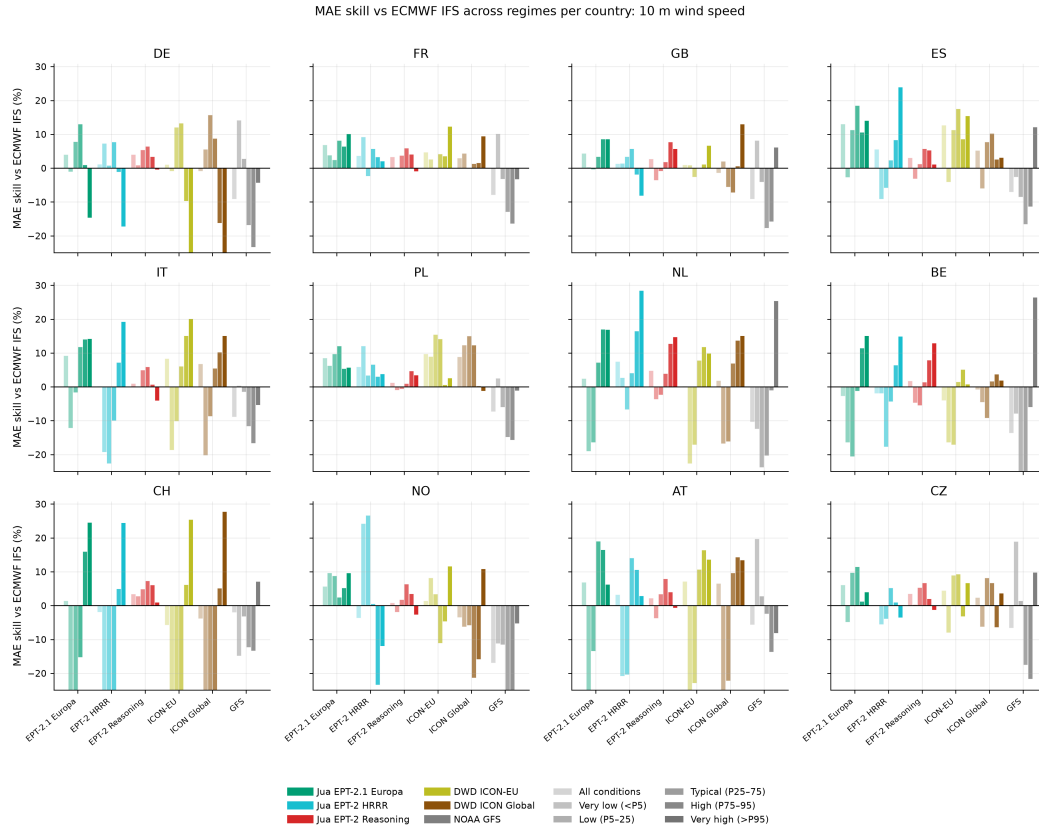


Figure S12: Per-country MAE skill by ERA5 1991–2020 climatological regime, 10 m wind, March–June 2026, pooled 1 to 48 h (hourly). Mixed regional/global model set.

1350
1351
1352
1353
1354
1355
1356
1357
1358
1359
1360
1361
1362
1363
1364
1365
1366
1367
1368
1369
1370
1371
1372
1373
1374
1375
1376
1377
1378
1379
1380
1381
1382
1383
1384
1385
1386
1387
1388
1389
1390
1391
1392
1393
1394
1395
1396
1397
1398
1399
1400
1401
1402
1403

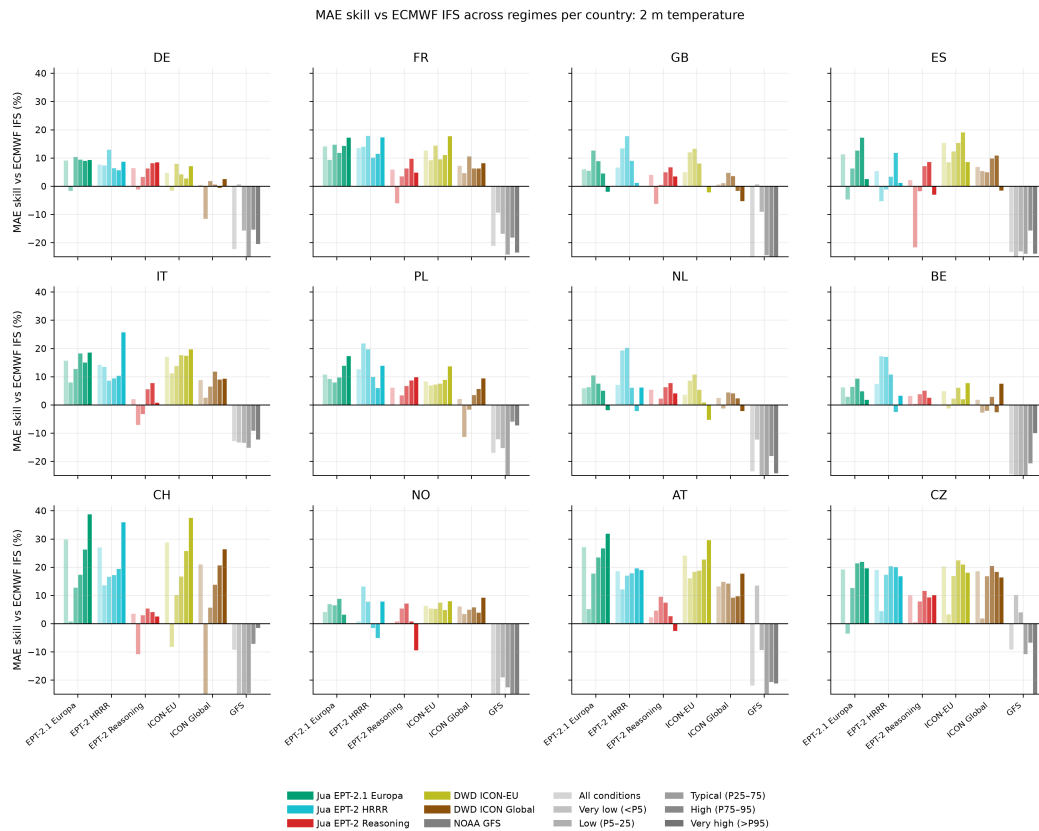


Figure S13: As Figure S12, for 2 m temperature.

1404
1405
1406
1407
1408
1409
1410
1411
1412
1413
1414
1415
1416
1417
1418
1419
1420
1421
1422
1423
1424
1425
1426
1427
1428
1429
1430
1431
1432
1433
1434
1435
1436
1437
1438
1439
1440
1441
1442
1443
1444
1445
1446
1447
1448
1449
1450
1451
1452
1453
1454
1455
1456
1457

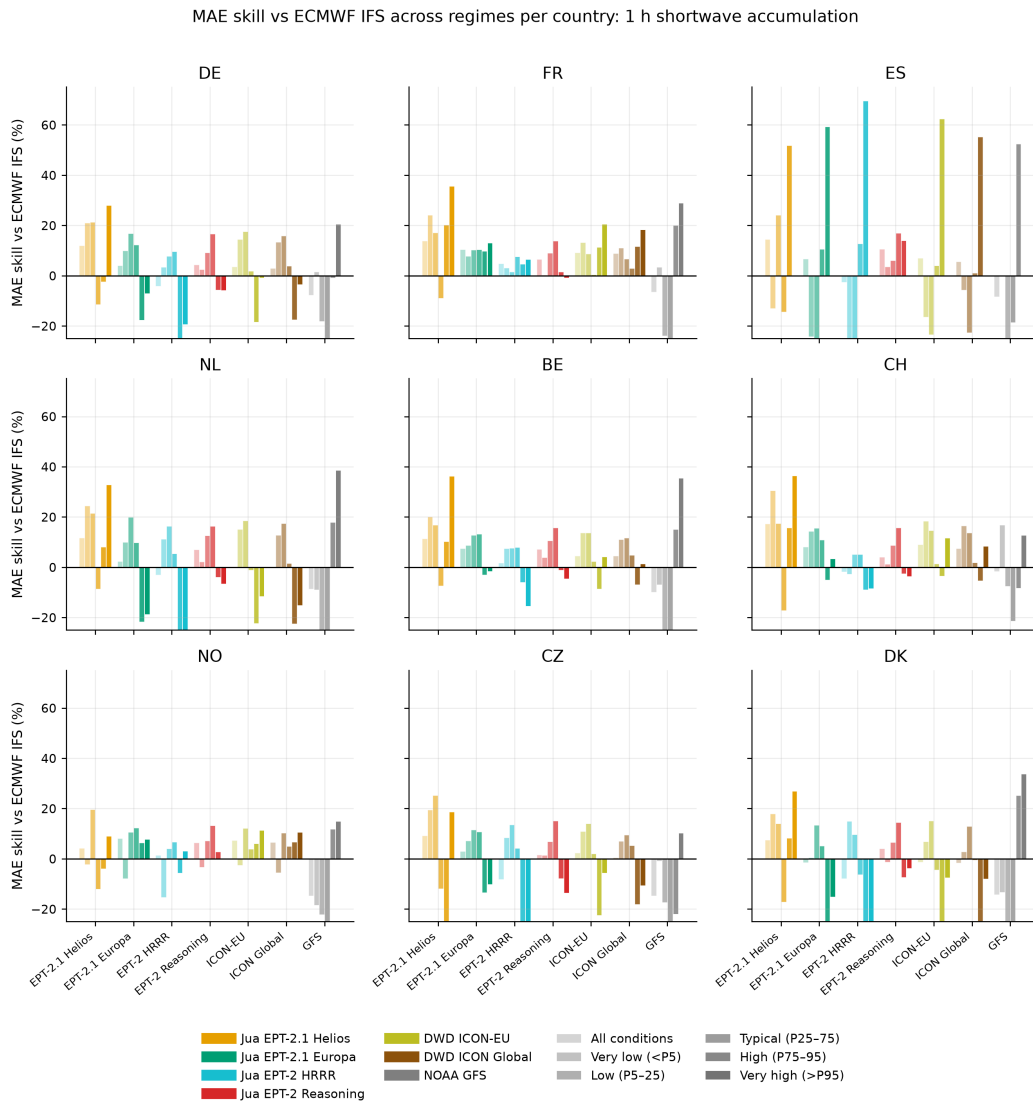


Figure S14: As Figure S12, for 1 h shortwave, with EPT-2.1 Helios added. GB, IT and PL have no solar station observations (Section 2.2).

1458
1459
1460
1461
1462
1463
1464
1465
1466
1467
1468
1469
1470
1471
1472
1473
1474
1475
1476
1477
1478
1479
1480
1481
1482
1483
1484
1485
1486
1487
1488
1489
1490
1491
1492
1493
1494
1495
1496
1497
1498
1499
1500
1501
1502
1503
1504
1505
1506
1507
1508
1509
1510
1511

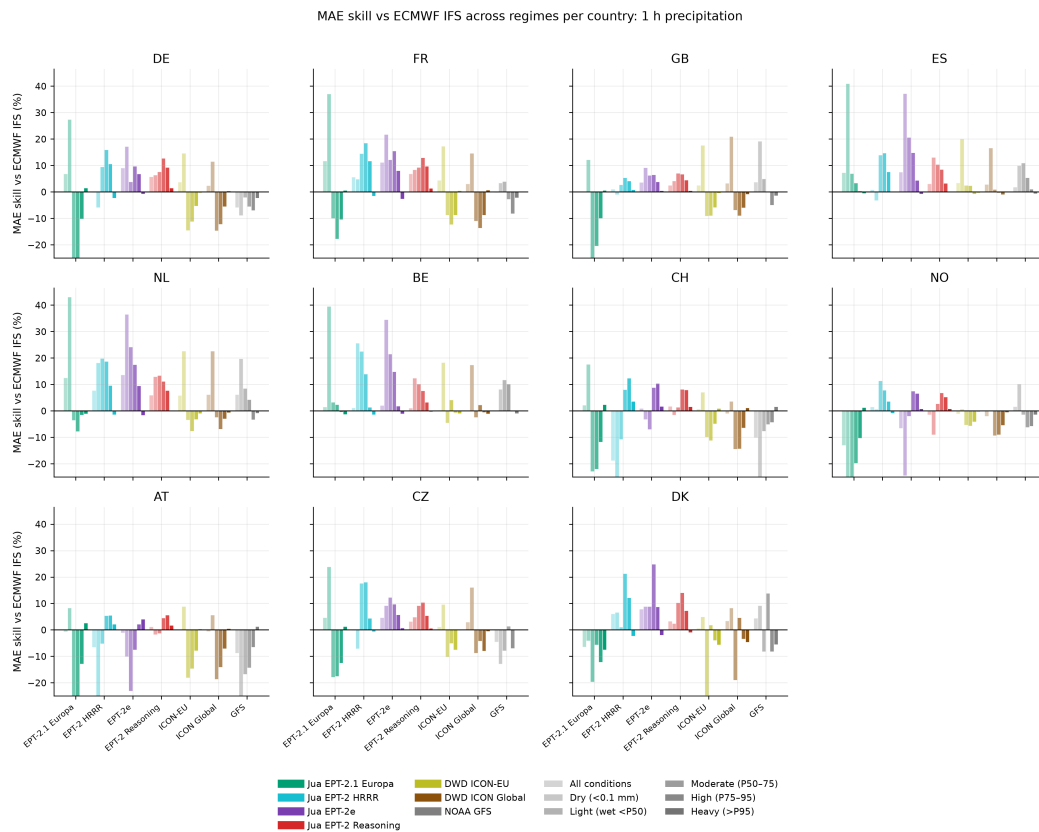


Figure S15: Per-country MAE skill by precipitation regime for the mixed regional/global model set. IT and PL have no gauges in the precipitation panel.

A.6 SOURCE-DEFINED EVENT CASE STUDIES

Table S25 lists the externally documented record-temperature events and European windstorms used for the case-study analysis. Event dates and affected countries are fixed from the source catalogue before forecast scoring. Within each footprint, forecasts are retained only where every Track A model and ECMWF IFS share the same station, valid time, initialisation, and lead; ensemble systems are evaluated through their means and the same causal bias correction as in the main analysis is applied.

Table S25: Source-defined event catalogue. The accompanying reproduction package records the source URLs and selection basis for every event.

Variable	Event	Dates	Affected countries
Temperature	Central European December record warmth	2025-12-07–2025-12-09	NL, CZ
Temperature	Late-February European record warmth	2026-02-25–2026-02-27	FR, DE, ES, CZ
Temperature	Early-April record warm spell	2026-04-05–2026-04-08	FR, GB
Temperature	Western European May heatwave	2026-05-21–2026-05-30	FR, GB
Temperature	Late-June European record heatwave	2026-06-21–2026-06-28	DE, FR, GB, ES, PL, NL, CH, AT, CZ, DK
Wind	North Sea autumn storm	2025-09-15–2025-09-16	NL
Wind	Storm Amy	2025-10-03–2025-10-05	GB, FR, BE, NL, NO, DK, PL
Wind	Storm Benjamin	2025-10-23–2025-10-24	FR, NL, CH, GB
Wind	Storm Goretti	2026-01-08–2026-01-09	GB, FR, DE, NL
Wind	Storm Nils	2026-02-11–2026-02-13	FR, ES
Wind	Storm Marlis	2026-03-25–2026-03-26	CH, DE
Wind	Storm Deborah	2026-03-26–2026-03-27	FR, IT

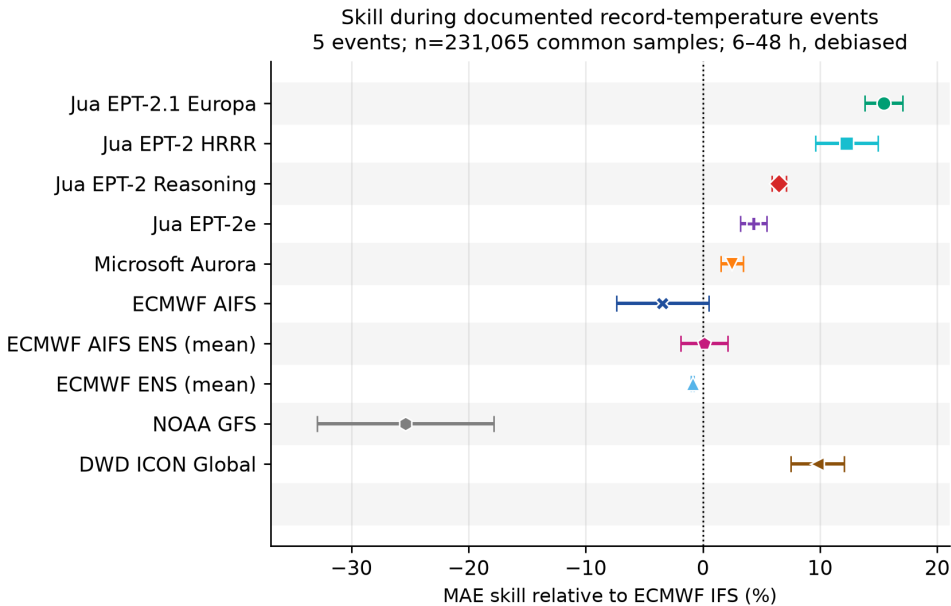


Figure S16: MAE skill relative to ECMWF IFS over the source-defined record-temperature event footprints, aggregated over 6–48 h after bias correction. Every model is evaluated on the same five events, eight leads, and 231,065 station/valid-time/lead samples. Error bars show one leave-one-event-out jackknife SE.

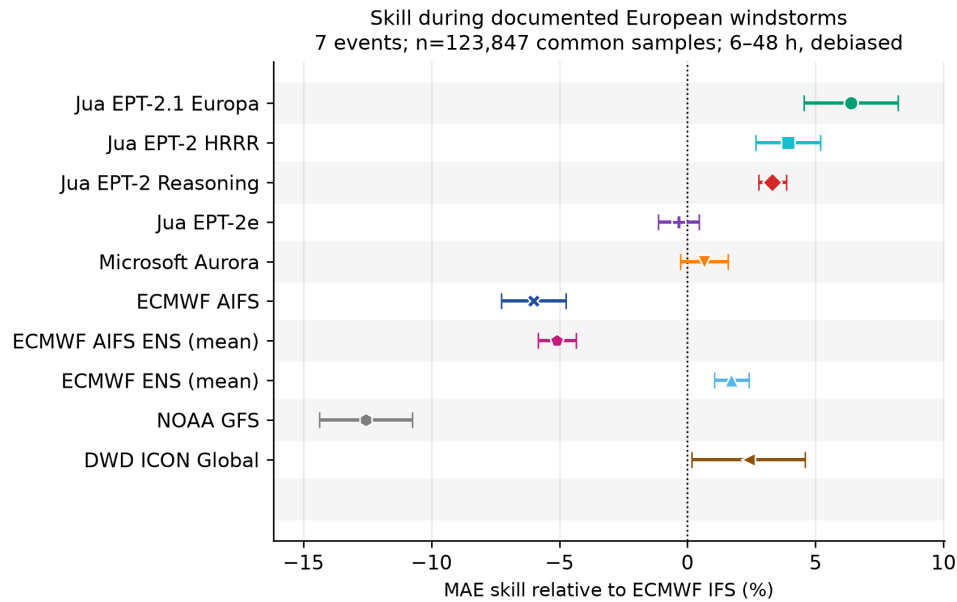


Figure S17: As Figure S16, for the seven externally documented European windstorms. Every model is evaluated on the same eight leads and 123,847 station/valid-time/lead samples.

Table S26: Event-case MAE skill (%) relative to ECMWF IFS. Parentheses give leave-one-event-out jackknife SE. Temperature uses five events and 231,065 common samples; wind uses seven events and 123,847 common samples.

Model	Temperature events	Windstorms
Jua EPT-2.1 Europa	+15.4 (1.6)	+6.4 (1.8)
Jua EPT-2 HRRR	+12.3 (2.7)	+3.9 (1.3)
Jua EPT-2 Reasoning	+6.5 (0.6)	+3.3 (0.5)
Jua EPT-2e	+4.3 (1.1)	-0.3 (0.8)
Microsoft Aurora	+2.5 (0.9)	+0.7 (0.9)
ECMWF AIFS	-3.4 (3.9)	-6.0 (1.3)
ECMWF AIFS ENS (mean)	+0.1 (2.0)	-5.1 (0.7)
ECMWF ENS (mean)	-0.9 (0.1)	+1.7 (0.7)
NOAA GFS	-25.4 (7.5)	-12.6 (1.8)
DWD ICON Global	+9.8 (2.3)	+2.4 (2.2)